HELLENIC MEDITERRANEAN UNIVERSITY
Department of Music Technology and Acoustics

DOCTORAL RESEARCH PROPOSAL

The Byzantine Melodic Formula as a Computational Unit

Symbolic Representation, Historical Transmission and Cognitive Function

Candidate	Georgios Spyridakis
Research fields	Symbolic music information retrieval, representation learning, computational musicology and music cognition
Musical scope	Melodic formulae in papadic Cherubic Hymn settings and Communion chants in New Method notation
Duration	48 months
Supporting documents	Detailed technical and musicological supporting documents are available separately
Repository	`github.com/GeorgeSpy/byz-position-phd-study`
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Abstract

The melodic formula (*thesis*, plural *theseis*) is a recurring, variable-length formation in Byzantine chant whose internal organisation supports recognition across different melodic realisations. This project studies formulae as computational units in papadic Cherubic Hymn settings and Communion chants written in the analytical New Method (1814). The tradition combines a historical theory of formulae with surviving parallel analytical renderings of the same older passages, known as *exegeses*.

Documented musicological knowledge defines the units learned, the relations used in training and the evidence used in evaluation. Formula families have explicit, versioned membership specifications validated independently. The starting point is the *Papadic Formulae* catalogue of St Anthony's Monastery, which draws on 758 compositions. Correspondence between parallel exegeses by Gregorios the Protopsaltes and Chourmouzios the Chartophylax is established through source criticism, separately from family membership.

The first route introduces joint formula segmentation and family recognition in continuous chant, accommodating alternative acceptable boundaries, unknown families and calibrated abstention. Baselines capture local boundary cues, melodic repetition and unsupervised segmentation. The central experiment compares two sources of positive training relations under a shared encoder architecture, common family labels and matched exposure and difficulty: same-family occurrences in different works and verified parallel exegeses of the same older passage. It tests how the more specific historical relation shapes representations of within-family variation. The independent evaluation criterion is agreement with the timing variability of contemporary chanters performing the same written passages; performance measurements are reserved for evaluation.

The second route models exegesis as a historical transformation channel. It tests whether patterns of expansion and contraction transfer to families withheld from training, and whether analytical exegesis supports probabilistic recovery of the older notational sign recorded in a specific manuscript witness. Recovery compares candidates with the same intervallic reading (metrophony) against known exegetical rules and a simple empirical comparator. Independent, blind palaeographic readings establish the reference labels. A pilot study measures the additional predictive value of a known family for its specific realisation, given modal and prosodic context.

Independent expert judgements validate the family lexicon, while listening experiments examine melodic expectation and perceptual categorisation. Analysis of complete works by Nikolaos Protopsaltes of Smyrna distinguishes family selection from internal realisation against a chronologically controlled reference corpus.

For machine learning, the contribution is a controlled test of how supervision provenance shapes representations of musical identity and variation. For symbolic music information retrieval (MIR), it is a musicologically specified task for structured recognition and a benchmark with traceable provenance. For Byzantine musicology, it is a validated family lexicon and quantitative methods for the morphology of exegesis, historical recovery and the family–realisation relationship. A source inventory in the first six months sets the scope of quantitative historical tests through predefined adequacy criteria.

Keywords: Byzantine chant, New Method, melodic formula, exegesis, symbolic music information retrieval, melodic segmentation and categorisation, representation learning, inverse historical recovery, musicology-informed machine learning.

1. Introduction: problem and motivation

1.1 From neume to complete work: what unit does a model learn?

The choice of musical unit determines what a computational model can recognise. Individual musical events—the elementary encoded melodic movements—describe the succession of pitches, whereas recurring formations are recognised in larger units of variable length. In symbolic music information retrieval (MIR), the input is encoded musical notation and the output may consist of boundaries, related occurrences or family categories; the choice of unit therefore shapes every subsequent inference. The relationship between statistical and musicological segmentation has already been examined in Gregorian chant (V. Lanz and J. Hajič Jr., 2025). This project addresses two related questions: how to locate a musically meaningful formation of variable length, and what evidence allows its identity to be learned across different realisations.

Byzantine chant offers a setting in which both questions can be studied within a tradition that has a historical theory of melodic formations and surviving alternative renderings of them. The **New Method**, developed from 1814 through the work of Chrysanthos of Madytos, Gregorios the Protopsaltes and Chourmouzios the Chartophylax, writes melody using *neumes*: signs that indicate relative intervallic movements and combine with temporal indications of relative duration and qualitative indications of melodic delivery. Their interpretation depends on the *echos*, the modal framework of pitches and cadences within the system of eight modes, the *oktoechos*. Reading the quantitative signs as a succession of intervals is known as *metrophony*. Metrophony provides the intervallic skeleton on which a fuller melodic realisation is built.

In the papadic repertory, extended melismatic elaboration distributes many musical events over a single syllable. This study focuses on settings of the Cherubic Hymn and on Communion chants, the hymns sung during communion. A **melodic formula** (*thesis*; plural *theseis*) is a recurring melodic formation of variable length whose internal organisation allows it to be recognised across works and sources. An *occurrence* is its localised presence in a particular passage; a *realisation* is the melodic form it takes there, in relation to mode, scale degree, syllabic distribution and the degree of notational detail. A *formula family* brings together occurrences that satisfy the same musicological specification. The specification states required relations for membership, supporting relations that strengthen the attribution, and exclusion criteria, allowing alternative sufficient combinations of features. A formula's historical name and the genealogical relationships between its sources are documented separately from this modern analytical category.

Two cases illustrate why diagnostic criteria are needed alongside melodic similarity. K. Ch. Karagounis (2003, p. 788) examines passages with related metrophony that are distinguished in older notation by signs such as the antikenoma, lygisma and heteron parakalesma. These passages provide *candidate hard negative pairs*: treating them as negative pairs requires an independent judgement that they belong to different families. Similarly, I. Arvanitis (2010, vol. A, §5.2, pp. 136–142) examines the recognisability of formulae despite changes in syllabic distribution and accentuation, while S. Karas's table of the parakletike in different modes (in Karagounis, 2003, p. 812) compares different realisations. Such cases can provide *hard positive pairs* once shared family membership is confirmed. Study 1 (M1) estimates boundaries and family jointly, because recognition depends on the extent of a segment, and its extent depends on what is recognised.

Figure 1 traces musical organisation from neumes to the complete work. The formula occupies an intermediate morphological level, where local notational information forms patterns that can be compared across works. The family denotes an identity relation between occurrences and is therefore shown below the formula, outside the temporal sequence.

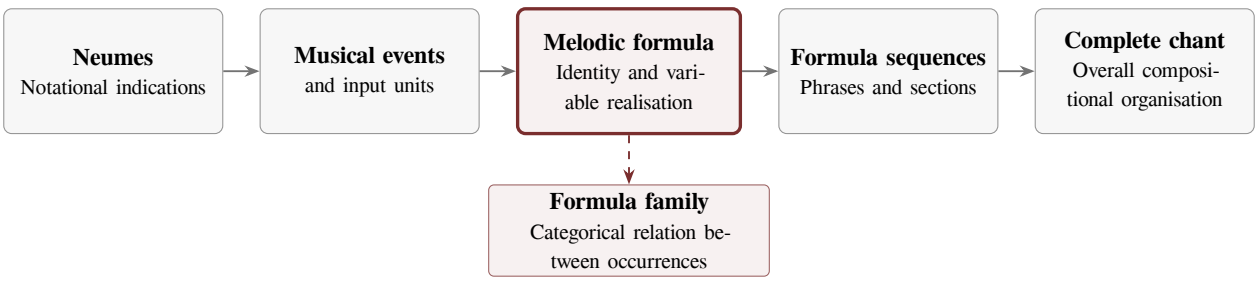


Figure 1. From neumatic notation to the structure of a complete chant: the melodic formula as an intermediate morphological and computational unit. Solid arrows indicate levels of organisation; the dashed relation indicates that the family connects occurrences and realisations outside the temporal sequence. M1 and M2 operate at formula level, M3 validates families with experts and listeners, and M5 moves to formula sequences and complete chants.

Why formulae and families. The formula is chosen because it recurs across works and supports comparison at corpus scale. Historical and morphological research uses it to describe melodic organisation (§2.1), while the effect of an individual sign depends on its context (Chrysanthos, 1832, §408). Analysis by family allows repetition and variation to be measured separately. Related approaches include tune families in oral tradition (A. Volk and P. van Kranenburg, 2012) and the distinction between core tones and surface formulae in galant schemata (G. Rabinovitch and A. Carter-Enyi, 2024). The cognitive role of the family in listening is tested empirically in Study 3 (M3).

Figure 2 shows the material from which the project starts, taken from the online *Papadic Formulae* catalogue of St Anthony's Greek Orthodox Monastery, Arizona. The catalogue organises examples by mode and prosodic pattern—a sequence of unstressed and stressed syllables—providing a resource for proposing and comparing candidate families. The prosodic code is a retrieval key; family attribution is established through independent musicological analysis.

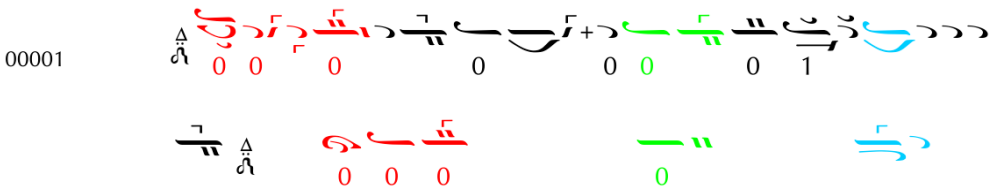


Figure 2. Notated variants of a melodic type from St Anthony's Monastery's *Papadic Formulae* catalogue. `00001` is a *five-syllable prosodic pattern code* (0: unstressed; 1: stressed), rather than a formula or family identifier. The catalogue grouping supports comparison of the forms; family membership is examined independently using musicological diagnostic criteria.

The studies follow this organisation. M1 selects the representation and jointly estimates boundaries and family; Study 2 (M2) examines the relations from which identity is learned; M3 validates families with experts and listeners; Study 4 (M4) measures contemporary performance variability and provides the independent criterion for M2; and Study 5 (M5) analyses complete works.

1.2 Exegesis: a supervision relation and a trace of historical transformation

The more detailed notational rendering of older chants is known as **exegesis**. The increasingly analytical notation of the eighteenth century prepared the way for its systematic recording in the New Method (G. Th. Stathis, 1978). Analytical renderings by Gregorios and Chourmouzios survive for some older passages. Each pair requires critical examination of the sources: which older passage is being explained, how the syllables correspond, and what dependence through copying or a shared working source connects the renderings.

In the **first route**, a verified pair provides *relational supervision*: it tells the model which realisations are historically connected. Eligibility requires a documented exegetical relationship, passage homology—correspondence to the same specific older passage—and an independent judgement of shared family membership. The experiment tests what this more specific relation contributes beyond a general family label. The observed material consists of surviving written exegeses; their relationship to historical performance is examined in the historiography of exegesis (Stathis, 1978; A. Lingas, 2003; Arvanitis, 2022).

In the **second route**, an exegesis is an observable outcome of historical transformation. Different older notations may share the same metrophony. As a schematic example, the oligon and petaste can both indicate an ascent of one step while serving different notational functions. Selected aphonic great hypostases, which have no independent quantitative movement, likewise combine with a shared quantitative basis and are associated with different melodic elaborations. Great hypostases are signs or sign complexes in older notation, with contextual and mnemonic functions in formula recognition (M. Alexandru, 2012). An analytical rendering may preserve predictable traces of these distinctions. The passages and candidate historical labels are selected after direct examination of the witness—the specific primary document—separately for Middle Byzantine notation and for the later, partly exegetical notations preceding 1814. Passages in which an older sign survives unchanged in the New Method are examined separately as controls.

The passages discussed by Karagounis (2003, p. 788) thus provide candidate material for two distinct tests: family discrimination, where independently confirmed, and prediction of the older notational label from the exegesis. The two routes share a source registry, alignments and splitting rules, while retaining different targets and held-out test sets (§4.8).

1.3 Key terms

Table 1 records the distinctions needed for source criticism and annotation. Formula, occurrence and realisation are defined in §1.1.

Table 1. Key terms and their role in the project

Term	Definition in this project
Historical formula	A melodic formation as recognised or named in a particular source. Its correspondence to a modern family is documented separately for each period.
Family and specification	An analytical category (§1.1) and a versioned membership specification. Alternative sufficient combinations are allowed; historical relatedness and cognitive function are tested separately.
Metrophonic class	Historically eligible notations that yield the same intervallic reading of the quantitative signs.
Older witness	The specific manuscript or primary document containing the older passage. Its relationship to the exegete's working source is recorded separately.
Exegetical relationship	A documented connection between an analytical rendering and a specific older passage. Family membership is judged independently.
Passage homology	Correspondence of two exegeses to the same older passage, established from the poetic text, syllabic alignment and location within the work.
Performance	A chanter's vocal rendering of a written realisation, with its own timing and pitch shaping.

In machine learning, *supervision* is the information from which a model learns; a *positive pair* indicates that two examples are connected by the relation under study. An *encoder* converts symbolic input into a numerical representation whose distances are used for comparison. Parameters are estimated on *training data*, model choices are made on *development data*, and a separate *held-out test set* evaluates the finalised choices. In this project, datasets are split by work and by groups of related sources to maintain their independence.

1.4 Research idea and contribution

The project rests on one idea: documented musicological knowledge defines the units a model learns, its training relations and the independent evidence used to evaluate it. The formula is the level at which structural recognition, historical variation, probabilistic realisation and the characterisation of complete works meet. This idea yields four testable contributions, detailed in §8.

C1—structured recognition. The selected comparators address complementary aspects of structure: local discontinuity, melodic repetition, statistical segmentation and categorisation (§2.2). M1 jointly estimates segments and families in continuous, partially annotated works, allowing alternative acceptable boundaries, an open set of families and abstention. Membership is assessed through independently validated diagnostic criteria; historically difficult pairs test whether the model distinguishes related metrophony from shared family membership.

C2a—history as a source of supervision. Shared family membership tells the model that two occurrences are "of the same kind"; two exegeses of the same passage indicate "the same object", rendered analytically by two musicians. M2 compares these relations within the same architecture, with shared family labels, and evaluates them against timing variability in independent contemporary performances (M4), held out from training. Contemporary performance provides a distinct level of evidence about formula structure; its agreement with historical written variation is the hypothesis tested by M2.

C2b—exegesis as a historical channel. The historical source itself becomes a prediction target. Attested Transformation Learning (ATL) learns how one exegesis transforms into another and tests whether these regularities transfer to new families. Inverse historical recovery (ATL-INV) reverses the channel: from an analytical rendering, it estimates which sign the older witness recorded, among candidates sharing the same metrophony, and is compared with known exegetical rules. Shared metrophony defines a controlled problem: the additional predictive value of analytical notation for distinguishing the older label is measured under common contextual information and predefined comparators. The pilot M_{thesis} models the possible realisation of a known family; the three tasks predict, respectively, the other exegesis, the older label and the realisation.

C3—musicological and cognitive knowledge. Families are validated through blind judgements by independent experts (M3a). Their role in listening is examined through melodic expectation (M3c) and categorisation with controlled surface similarity (M3b). M5 applies the validated lexicon to complete works and distinguishes composition through the choice of familiar families from composition through different realisations of those families.

The main deliverables are organised into priority tiers 1–3 (Table 4), with the minimum performance criterion included in the core of M2. The historical strands deliver a source inventory and documentation; blind readings

and alignments are added for accessible material that can be matched. Quantitative tests are activated according to the adequacy of independent data (§4.2). The methodological contribution connects structured prediction and probabilistic learning with a new recognition task, a controlled comparison of historical supervision and a quantitative framework for historical recovery.

2. Theoretical framework and literature review

2.1 Formulae and exegesis in musicological research

Historical theory recognises the formula as a unit between the sign and the chant. Manuel Chrysaphes, a fifteenth-century lampadarios and theorist, connects the formula with the combination of signs that constitutes melody (ed. D. E. Conomos, 1985, pp. 40–41). Gabriel Hieromonachos illuminates it through the grammatical analogy of the word (ed. C. Hannick and G. Wolfram, 1985, p. 72). Chrysanthos (1832, §§400, 408) describes transmission and the dependence of a sign's effect on its context. These sources document historical conceptions of melodic organisation; the project develops operational family definitions from their critical examination. The correspondence between a historical name and a present-day analytical family will be documented by period and notational phase.

Modern morphological research has developed this recognition into an analytical practice. A. E. Alygizakis (1985, pp. 81–82) describes formulae as reusable melodic patterns that form phrases and complete chants. Arvanitis (2010, §5.2) examines the formula as a principle of melodic similarity across differences in notation, accentuation and syllabic distribution. C. Troelsgård (2011) and M. Alexandru with Troelsgård (2020, pp. 175–180) analyse the notational and modal functions of these formations. For the papadic repertory, studies by Stathis (1978, 1987), Karagounis (2003, 2005), N. Gheorghiță (2010) and D. P. Spanoudakis (2017) document named formulae, recurrences, formula chains and their exegeses. Analysing forty-two eighteenth-century Sunday Communion chants, Gheorghiță identifies four factors that alter a formula's character: its degree within the modal structure, genus, compositional idiom and the written colour of the cheironomic sign (red or black). He also compares older and more analytical notations of the same chant retrospectively. Comparing exegeses is thus an established musicological method, and Karagounis himself (2003, p. 722) identifies detailed comparison of this kind as the subject of a separate morphological investigation. The project develops this practice into a reproducible procedure at corpus scale, retaining the individual source judgements that establish its musicological validity.

The study of older signs and exegesis has its own historiography, with each work supporting a particular part of the design. Stathis (1978) provides a foundational study of exegesis. The project distinguishes evidence from written analytical renderings from inferences about historical performance, drawing on the wider historiography (Lingas, 2003; Arvanitis, 2022). The theoretical treatises of Apostolos Konstas of Chios, studied by Th. K. Apostolopoulos (2002), and K. A. Psachos's (1917/1978) historical and palaeographic synthesis on Byzantine notation provide early systematic precedents for the study of exegesis. Candidate notational classes and known exegetical rules are inventoried from these works before the pairs are examined, to serve as a reference model for historical recovery; Psachos's interpretations are also considered as historiographical positions, with their criteria explicitly recorded. Arvanitis's research on rhythm and the exegesis of older notation (2010), and on the transcriptions of the Three Teachers (2022), defines the scope of inferences about sound from notation.

Two works frame the historical question of the second route. M. Alexandru's (2012) study of the great hypostases relates the great signs to formula recognition, mnemonic knowledge, gesture and contextual modes of melodic realisation; the great hypostases are therefore the priority for historical recovery. S. I. Karas (1982), working within a later theoretical framework, examines the signs and their effects and argues for the survival of qualitative distinctions between older signs through oral teaching and exegeses. The project gives this question a testable written form: does a distinction in older notation leave a recoverable trace in analytical notation? Material from the Karas school also informs the analytical training of the lexicon's editor (cf. the table on p. 812 of Karagounis); every such shared source is therefore recorded in the knowledge-role ledger (§4.1).

The tradition itself provides precedents for the difficulties of comparison. Gheorghiță (2010) cites Chrysanthos's recommendation to compare older and newer notations of the same chant to understand how it was sung, and describes formulae rendered ambiguously by the Three Teachers, such as the *epegerma*, as well as signs whose distinctive melodic elaboration is omitted in exegesis. Correspondence between an older passage and its exegesis can therefore be ambiguous or partial; the project registry records these cases as historical findings.

Digital encoding of older notation depends on the same reading decisions. E. De Luca, M. Alexandru and I. Fujinaga (2023) emphasise that reading a Middle Byzantine sign depends on neighbouring graphic relationships, chant genre, mode, melodic formula and text. Historical recovery is therefore organised into candidate classes with independently documented readings and correspondences. Eligibility of the historical label, passage alignment and family membership are assessed separately.

2.2 Symbolic music information retrieval: units, relatedness and transmission

Symbolic MIR has studied both problems posed by the formula—segmentation and categorisation—and demonstrated their interdependence. E. Cambouropoulos's work provides a foundational account of this relationship. The *Local Boundary Detection Model* (LBDM; 2001a) proposes boundaries from local discontinuities on the melodic surface and examines their relationship to expressive timing. Segmentation through melodic parallelism (2006) places boundaries at the beginnings and ends of significant repetitions, allowing repetition to contribute to segmentation. The *cue abstraction* model (2001b), with the *Unscramble* algorithm, and motivic analysis through melodic clustering (Cambouropoulos and Widmer, 2000) treat similarity and categorisation as interdependent processes: feature weights are recalculated from the categories that emerge, and similarity is always context-dependent (2001b; 2009).

The relationship between segmentation, repetition and categorisation has since received sustained attention in symbolic MIR, from the discovery of recurrent segmental patterns (D. Conklin and C. Anagnostopoulou, 2006) and retrieval of known melodic segments (B. Janssen et al., 2017) to statistical segmentation of liturgical chant (Lanz and Hajič, 2025). In most of these studies, categories either emerge from the data or are supplied as a search query. Byzantine chant offers another setting: musicologically defined families with explicit, independently validated specifications and documented correspondences to historical terminology, on which joint estimation of segments and categories can be studied. M1 therefore uses LBDM and segmentation through parallelism as interpretable comparators, showing when formula boundaries are explained by local discontinuity, when by repetition and when they require family knowledge. Each family specification explicitly states its diagnostic properties through required and supporting relations and alternative sufficient combinations. M3b tests the contribution of these properties to categorisation judgements while accounting for overall surface distance. A hierarchy of pitches, formulae, phrasal patterns and larger units has also been proposed for the cognitive organisation of contemporary chant (Mavromatis, 2019); here it leads to a specific test of boundaries and family recognition.

In Latin liturgical chant, B. Cornelissen et al. (2020a, 2020b) examined which segmentation units support mode classification and developed tools for large corpora. Lanz and Hajič (2025) used non-parametric Bayesian models of the Pitman–Yor family and examined the relationship between statistically induced segmentation of Gregorian melodies and the musicological concept of centonisation. This finding is directly relevant here: correspondence between a statistical unit and a unit recognised by the tradition is an empirical question, requiring systematic examination with independently defined families and a common prediction target.

Once a candidate segment has been located, its family membership remains to be determined. Research on tune families has shown that relatedness can be annotated through explicit criteria, while the diagnostic strength of features varies both between and within families (Volk and van Kranenburg, 2012). Rabinovitch and Carter-Enyi (2024) describe a comparable hierarchy for galant schemata, distinguishing core tones, *outlines for diminution* and surface formulae; their first musical example is a melodic family from Greek chant in the first mode. For the present project, a specification must therefore state where each family's identity lies: in its skeleton or, for some families, in the surface pattern itself. Established methods also exist for learning similarity metrics in tune retrieval (F. Karsdorp et al., 2019), while A. Holzapfel's doctoral thesis (2010) shows, in traditional repertories, that local similarity between subsegments and shared identity of a larger unit are separate judgements. These methods provide reference models for surface distance and family recognition.

Historical transmission has been studied through melody alignment (P. E. Savage et al., 2022) and phylogenetic methods in Gregorian chant (Hajič et al., 2023). In Byzantine music, P. Mavromatis (2005a, 2005b) developed probabilistic models of melodic generation and connected formulae with written and oral transmission. P. G. Polykarpidis (2023; P. Polykarpidis et al., 2022) developed a hierarchical digital representation of Middle Byzantine notation and comparative analysis of heirmological corpora, while M. Panteli and H. Purwins (2013) compared Chrysanthine theory with performance practice. GraphMuse (E. Karystinaios and G. Widmer, 2024) provides a general graph infrastructure for symbolic music, representing information at several structural levels. The proposed project connects joint segmentation and family recognition of New Method formulae with controlled historical relational supervision and prediction of the notation recorded in a specific witness.

2.3 Representation learning: where positive pairs come from

Segmentation and retrieval define the task; learning identity also requires a decision about supervision. In metric learning, which estimates distances between examples, and in contrastive methods, which contrast positive and negative pairs, representations are shaped by relations between examples, and the members of each positive pair are encouraged to move closer. The attraction term in M2 uses positive relations and is defined separately from a full contrastive objective. Pairs are usually constructed either as artificial variants of the same example through researcher-selected augmentations (T. Chen et al., 2020), or as examples sharing a category label (P. Khosla et al., 2020). In music, metric learning for tune retrieval has used shared family membership as a source of positive pairs

(Karsdorp et al., 2019). Even when transformations are learned from data, they are selected within predefined parametric families, such as geometric image transformations (G. Benton et al., 2020). The source of positive pairs is therefore an experimental variable in its own right.

Shared category membership connects occurrences with different histories and internal forms, providing a clear basis for comparing sources of supervision. Two lines of research come closer to the present question. The first constructs positive pairs from the identity of the underlying object and shows improved generalisation to new environments (D. Mahajan et al., 2021). The second transfers the ways in which members of one category vary to new categories with few examples (B. Hariharan and R. Girshick, 2017). In both, the "same object" relation is supplied by the data collection process or inferred by the model; this project documents it from an independent historical source. These precedents correspond to two different uses of exegesis: in M2, historical evidence defines the supervision graph, with occurrences as nodes and positive pairs as edges; in ATL, attested changes become targets of prediction and transfer.

Formula segmentation additionally requires methods that score entire segments of variable length. In structured prediction, semi-Markov conditional random fields address precisely this problem (S. Sarawagi and W. W. Cohen, 2004), and training from incomplete annotations or under constraints has established precedents (Y. Tsuboi et al., 2008; S. Papay et al., 2022). For melodic change, musical edit distance provides a basic framework (M. Mongeau and D. Sankoff, 1990), with *fragmentation* and *consolidation* operations in which one note corresponds to several, and with costs estimated from examples (E. S. Ristad and P. N. Yianilos, 1998). These operations have been adapted to matching melodic variants under pitch, rhythm and parallelism constraints (M. Giraud, K. Déguernel and E. Cambouropoulos, 2014). Exegesis often takes this form: a sign or complex in older notation corresponds to several movements in analytical notation, while two exegeses of the same passage differ in expansions and contractions. Each model choice thus addresses a specific musical difficulty: uncertain boundaries, selective supervision or changes in length.

2.4 Inverse modelling and information accessibility

The second route treats exegesis as a directed probabilistic channel from older to analytical notation, and recovery as inversion of that channel. Direct methodological precedents include noisy-channel models for back-transliteration (K. Knight and J. Graehl, 1998), statistical language decipherment (B. Snyder, R. Barzilay and K. Knight, 2010) and probabilistic reconstruction of ancient languages (A. Bouchard-Côté et al., 2013). Section 8.3 explains how the project's historical recovery differs from these approaches. A related musical precedent is the use of dual-notation records to decipher Znamenny notation (V. D. Gusev et al., 2020).

Whether a learned representation retains information about an external property has been studied through capacity-controlled probing classifiers and control tasks (J. Hewitt and P. Liang, 2019), and through minimum description length criteria (E. Voita and I. Titov, 2020). For a fixed encoder that computes its representation from a given input, the data processing inequality sets the corresponding information bound (T. Pimentel et al., 2020). The test focuses on *ease of access* to information: what a simple classifier can read out. This distinction determines the interpretation of the historical route's accessibility test (stage R4, §4.8).

2.5 Informed machine learning and evaluation integrity

Musicological theory supplies more than labels: it defines units, permissible musical relations and the evidence against which they can be tested. Incorporating such knowledge falls within informed machine learning, which classifies where prior knowledge comes from, how it is represented and where it enters learning (L. von Rueden et al., 2023). In the physical sciences, incorporating known laws into model structure, losses or data has developed into a distinct field (G. E. Karniadakis et al., 2021); in MIR, model-based deep learning draws on musical and acoustic knowledge to improve interpretability, testability and efficiency (G. Richard et al., 2024).

The project's *musicology-informed machine learning* develops this approach for historical musicology. Its specific contribution is to give historically documented knowledge distinct roles at different points: defining the unit, imposing structural constraints, supplying relational supervision, serving as a prediction target and providing an independent criterion. Family categories are formulated as revisable musicological hypotheses. Their validity is tested using separate evidence for definition, development and independent evaluation. Research on leakage demonstrates how readily evaluation is distorted when modelling decisions use data that are later used for testing (S. Kapoor and A. Narayanan, 2023); in MIR, B. L. Sturm (2014) showed that high performance can rely on features unrelated to the musical concept supposedly measured. The knowledge-role ledger (§4.1) turns this principle into an auditable part of the method.

2.6 Listening, expectation and performance

Computational recognition of a family gains independent cognitive significance when it predicts how people listen and perform. One relevant line of research concerns the organisation of tune families in listening (Volk and van Kranenburg, 2012) and melodic identity under surface transformations (A. Schindler et al., 2013). Another connects probabilistic musical expectation with statistical learning of style (M. T. Pearce, 2018), boundary perception (Pearce et al., 2010; N. C. Hansen et al., 2021) and expressive timing (B. Gingras et al., 2016). Hierarchical musical syntax is examined at behavioural and neural levels by S. Koelsch et al. (2013) and S. A. Herff et al. (2021). G. M. Di Liberto et al. (2020) and P. Kern et al. (2022) document neural responses to melodic expectation and short-term predictions.

Mavromatis (2005b) already posed the cognitive question for chant: whether formulae function as units of a chanter's musical knowledge. This project turns it into two testable predictions for listening (§3.4): that family information explains judgements of unexpected continuation beyond individual events, and organises relatedness judgements beyond surface distance. Performance constitutes a different level of observation. The relationship between contemporary chanters' variability and historical written variants supplies the specific prediction against which the main M2 comparison is assessed.

2.7 Why this question is timely

The development of unified corpora and code libraries is expanding reproducible research on liturgical chant (A. Dvořáková et al., 2026). At the same time, evaluations of foundation models across musical traditions reveal uneven transfer of representations (C. Papaioannou et al., 2025). The proposed project examines a specific alternative source of learning for corpora with limited digital resources: carefully documented musicological and historical relations between occurrences. Its value is measured on new works and sources, with known collection costs and independent criteria.

3. Research gap, aim, questions and hypotheses

The review identified two mature areas: musicological analysis of formulae and exegeses, and computational segmentation and similarity learning. The research gap concerns their experimental integration, with separate evidence for definition, training and evaluation.

3.1 Research gap

The first open issue concerns musicological definition. Knowledge of formulae rests mainly on qualitative case studies and manual comparisons between works. The project builds a reproducible framework that turns historical names and published morphological observations into family specifications that independent experts can apply to new passages. It distinguishes sources of variation: the underlying formula, modal and textual context, the specific historical passage, the exegete and contemporary performance. This makes a question running through research on exegesis measurable (cf. Stathis, 1978; Lingas, 2003; Spanoudakis, 2017): what analytical notation preserves of a formula's identity, and how variation relates to exegete, source and musical context.

The second issue is computational. Inducing recurrent segments and recognising musicological units have different evaluation targets (Lanz and Hajič, 2025), while categorisation models often take segments as given (§2.2). The nature of the formula calls for a task that connects the two: locating and recognising musicologically defined units of variable length within continuous chants, with acceptable alternative boundaries where analysis supports several readings, an open set of families and abstention when evidence is insufficient. The relations from which identity is learned also remain open. The proposed comparison measures what a historically documented "same underlying object" relation, such as parallel exegeses of the same passage, adds beyond a "shared category" relation, using a common architecture, shared family labels, an equal budget and matched exposure, difficulty and pair-graph structure over their feasible common support. A separate analysis tests when independently validated diagnostic knowledge benefits learning through structured use. Where expert knowledge both defines labels and is itself tested, evaluation also requires an explicit record of the role of each piece of evidence.

The third issue is historical. Comparative examination of older notations and exegeses has extensive musicological documentation, while computational linguistics offers probabilistic models for reversing historical transformations (§2.4). The project connects these with independently established palaeographic reference labels. The proposed test quantifies which older notational choice can be recovered from one or two exegeses among candidates with shared metrophony, how far recovery improves on known rules, whether attested transformations transfer to new families, and how predictable a realisation becomes when its family is known.

3.2 Aim and objectives

The project aims to develop and independently evaluate a computational framework for the musical identity of formulae and the information preserved by historically documented exegeses. The framework combines structured recognition, controlled comparison of supervision sources, musicology-informed learning and probabilistic historical recovery. The aim is developed through six objectives:

S1—Infrastructure and lexicon. A common symbolic layer with a provenance registry and a versioned lexicon of formula families (WP0, WP1).

S2—Representation and structural unit. Selection of the input representation and assessment of the formula as a unit of boundaries and families on held-out works and sources (M1).

S3—Historical supervision. Controlled estimation of the added value of pairs of parallel exegeses over shared-family pairs, and a factorial test of diagnostic knowledge (M2, A2).

S4—Independent validity. Validation of families through blind expert judgements, listening tasks and contemporary performance (M3–M4).

S5—Complete works. Characterisation of Nikolaos Protopsaltes of Smyrna's compositions against an explicit, chronologically controlled reference corpus (M5).

S6—Historical transformation, recoverability and realisation. Transfer of attested transformations, recoverability of older notation and its accessibility in M2 representations; a pilot measurement of realisation predictability (ATL, ATL-INV, R4, M_{thesis}), with scope determined by the inventory.

3.3 Research questions

Central question. *How is a melodic formula's identity recognised across its variation, and what do historically documented exegeses contribute to learning that identity? What information about older notation remains predictable from these exegeses?*

The first part concerns learning musical organisation; the second concerns the label recorded in a specific historical witness. They are developed into four main questions, forming the research core, and two supplementary questions activated by predefined adequacy criteria.

RQ1—Representation and unit (M1). Which input unit and which notational, temporal, modal and textual features best support boundaries and family distinctions? What does an explicit formula unit contribute beyond an event model and statistically induced segments?

RQ2—Provenance of supervision (M2). What does the exegetical relationship to the same passage add to learning within-family variation beyond shared family membership, when training is compared under common criteria? A separate subquestion examines when structured use of musicologically valid diagnostic relations improves learning.

RQ3—Independent validation (M3, M4). How reliably do experts apply the specifications? How do families contribute to melodic expectation and perceptual categorisation? How is contemporary performance variation organised? The minimum M4 component supplies the external criterion for RQ2.

RQ4—Complete works (M5). Do Nikolaos of Smyrna's compositions differ from the predefined reference corpus primarily in family choice or in the realisation of familiar formulae? Where data are adequate, how do their sequencing and functional roles differ?

RQ5—Exegesis as a historical channel. Do regularities of expansion and contraction transfer to families withheld from estimation? Which older labels sharing the same metrophony can be recovered beyond known rules, and what does a second exegesis add on the same passages? How accessible does historical information remain in the finalised representations?

RQ6—Predictability of realisation. How much additional predictive value does a known family provide beyond local, modal and prosodic context, and for which occurrences can family membership be determined reliably enough to test this?

3.4 Estimands, hypotheses and inferential status

The main M2 estimand. In the main experiment, two conditions train the same encoder (§4.4): B2 uses shared-family pairs, and EISE uses verified parallel exegeses of the same passage. For each model m, local states yield a *historical variation profile* across internal regions of the formula; the minimum M4 component provides the corresponding *performance profile* of between-chanter timing variability from recordings held out from training. Agreement S_m is the mean of within-family Spearman rank correlations, with equal family weights (§4.4), and the primary M2 contrast is

$$\Delta_{\text{hist}} = S_{\text{EISE}} - S_{\text{B2}}.$$

A positive value means that supervision from the same passage organises internal variation in closer agreement with regions of greater performance variability; a negative value means that shared-family supervision agrees more closely. The test is two-sided: the uncertainty interval from cluster bootstrap resampling of whole works and source groups is assessed against a smallest effect size of interest specified in the preregistration. A value near zero is interpreted alongside measurement reliability and representation diagnostics. The estimand has confirmatory status only over the scope supported jointly by the pair and performance-data adequacy criteria (§4.2). The difference $\Delta_{\text{learn}} = S_{\text{EISE}} - S_{\text{B0}}$, relative to a strong symbolic description without learning (B0: the same symbolic features of $R^\star$, by region), is a secondary descriptive estimand.

Confirmatory listening hypotheses. M3 tests two hypotheses as one family using Holm correction for multiplicity. Under the *hierarchical expectation hypothesis* (H_{pred}, M3c), the information content of the next formula family, $I_F = -\log_2 P$, positively explains listeners' judgements of how unexpected a continuation is, beyond the information content of individual events, I_{event}. Under the *categorical organisation hypothesis* (H_{cat}, M3b), listeners in triadic choice tasks organise realisations by family membership beyond their surface melodic and acoustic distances.

Historical estimands. The second route measures improvements in probabilistic prediction in bits, as differences in cross-entropy (mean negative log probability of the observed target) on held-out works. The primary ATL-INV estimand is $\Delta^{\text{rule}}_{\text{rec}}$, the gain from exegesis in predicting the older label relative to the strongest predefined comparator, supplemented by Δ_{rec} (gain over metrophony and context) and Δ_{2ex} (value of a second exegesis); ATL uses the transfer measure Δ_{ATL}; the pilot M_{thesis} uses Δ_{family}. Definitions appear in §4.8, following the musical question each measure addresses.

Table 2 links questions, studies, estimands, criteria and inferential status. The project distinguishes three inferential categories. Confirmatory tests comprise Δ_{hist}, subject to adequacy criteria, and the hypotheses H_{pred} and H_{cat}. M1 performance and the minimum M5 analysis are predefined descriptive quantities with uncertainty estimates. The status of historical and pilot studies is determined by the inventory: historical recovery (stage R3) is evaluated confirmatorily within its documented scope if the predefined adequacy criterion at months 12–18 is met.

Table 2. Questions, studies, estimands, criteria and inferential status in the research design

RQ	Study	Main measure	Criterion and comparator	Status
RQ1	M1	Boundaries (pooled F1, micro-F1) and family recall across twelve unit–feature conditions ($U \times F$); selection of $R^\star$; separate structural-unit comparison	Held-out works and source groups; event Transformer, Pitman–Yor, LBDM, parallelism; historically difficult examples (area under the ROC curve, AUROC)	Core (Tier 1)
RQ2	M2, A2, minimum M4	$\Delta_{\text{hist}} = S_{\text{EISE}} - S_{\text{B2}}$; Δ_{learn} separately; A2 2×2 effects	Timing variability of contemporary chanters, held out from training; B0/B1 as references	Core; confirmatory within the adequate scope
RQ3	M3a; M3c, M3b	Agreement on boundaries, features and membership; H_{pred}, H_{cat}	Blind expert judgements on new material; listeners with different experience; Holm correction	M3a: Tier 1; M3c/M3b: Tier 2
RQ4	M5	D_{fam}, D_{within}; D_{seq}, D_{func} subject to adequacy	Reference corpus Ω; independent musicological analysis; analysis of Nikolaos's Communion chant (Ch. Ioannidis, 2022)	Minimum: Tier 3; full: Tier 4
RQ5	ATL, ATL-INV (R0–R3), R4	Δ_{ATL}; $\Delta^{\text{rule}}_{\text{rec}}$ (primary), Δ_{rec}, Δ_{2ex}; accessibility of H	Independent witness reading; rule model; held-out works and families; R4: EISE versus B1 and symbolic input, B2 separately	Tier 4; R3 subject to months 12–18 criterion; R4 after M2
RQ6	M_{thesis} (P0–P2)	Δ_{family}; realisation typicality	Held-out works; family-free reference model; target hidden before coding	Tier 4 (pilot); neural decoder: Tier 5

4. Methodology

The methodology begins with shared infrastructure and adequacy criteria. Representation selection, the supervision comparison and independent validation follow; the historical route has separate targets and evaluation rules.

4.1 Two routes on shared infrastructure

The first step is a preparatory inventory (work package WP0): which witnesses, exegeses, occurrences and recordings can be connected through verifiable provenance? The two routes introduced in §1.4 use this infrastructure (Figure 3). In *Route A*, M1 selects the symbolic representation, M2 compares sources of relational supervision against an independent performance criterion from the minimum M4 component, M3 validates the lexicon and listening predictions, and M5 applies the validated lexicon to complete works. In *Route B*, ATL, ATL-INV and the pilot M_{thesis} predict, respectively, the other exegesis, the older label and the realisation. After M2 is finalised, stage R4 tests how accessible the historical label remains in the representations.

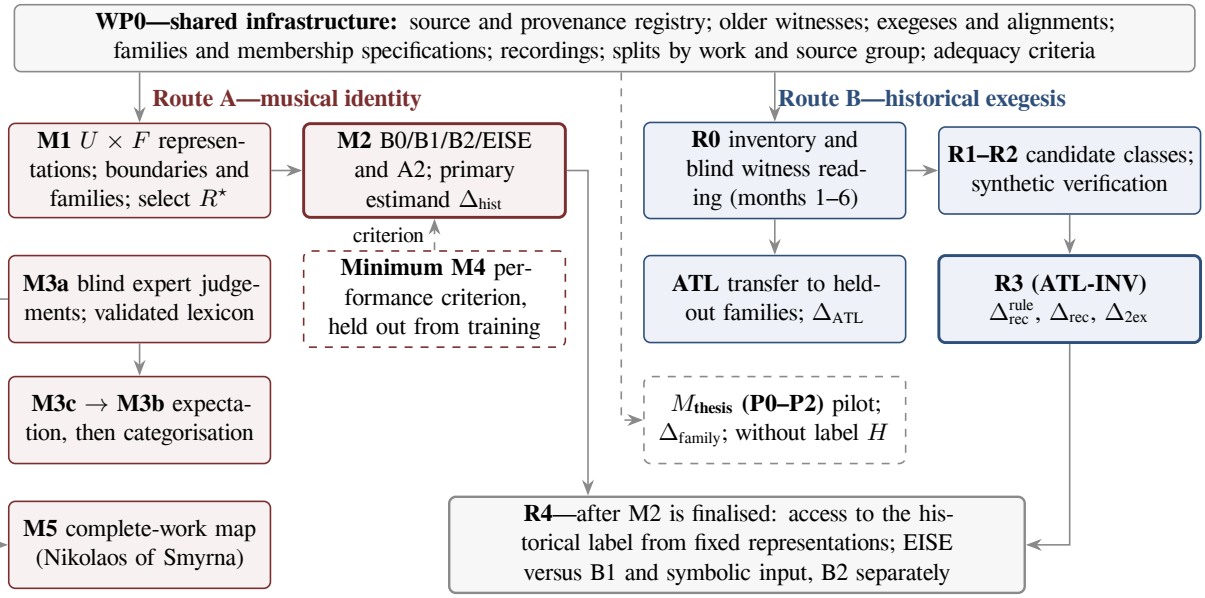


Figure 3. Two routes with shared infrastructure and distinct targets. R0 starts alongside WP0; the minimum M4 component delivers M2's criterion before Δ_{hist} is calculated; R4 follows finalisation of M2 and connects the routes while retaining their distinct targets. M5 uses the validated lexicon and finalised M2 models. The dashed edge from M4 denotes an external evaluation criterion; the edge to M_{thesis} denotes shared infrastructure, with no temporal dependence of the pilot study on M2.

Where musicological knowledge enters. Table 3 gives the operational form of the central idea in §1.4: musicological knowledge enters the design in different roles, each with its own validation.

Table 3. Where musicological knowledge enters and how it is tested

Role	Input	Validation	Study
Representation	Input units and feature layers (notational, temporal, modal, textual)	Comparison of twelve encodings on new works and sources	M1
Structural constraint	The formula as a variable-length segment with alternative boundaries, open-set recognition and abstention	Comparison with an event model and algorithmically induced segments	M1
Relational supervision	Positive pairs from shared family membership or parallel exegeses of the same passage	Matched B2–EISE comparison with an independent criterion	M2
Diagnostic relations	Required and supporting relations and exclusion criteria in the specifications	2×2 factorial design (valid/perturbed × flat/structured)	A2
Historical label and rules	Blind witness reading; exegetical rules recorded in advance	Label as prediction target; rules as a strong reference model	ATL-INV, R4
External criteria	Expert judgements, listening tasks, performance measurements	Evidence held out from training	M3, M4

Knowledge-role ledger. In historical musicology, family membership rules derive from theoretical sources and analytical judgements that are themselves open to scrutiny. Because the specifications serve both as a source of supervision and as hypotheses, each item in the registry has an explicit role: definition material, development material, independent validation material, or evidence used exclusively to establish a historical relation. Every specification

carries a version number, and examples used to formulate or revise it are excluded from later confirmation of that specification. Independence is protected at the levels of work, source, related-source group and documented transmission lineage. Four distinct judgements—computed similarity, family membership, exegetical relationship and genealogical relatedness of sources—are recorded in separate fields. Witness identifiers, interpretative relations, family labels and performance observations retain distinct roles, and written exegeses are treated as evidence of notation.

Priority tiers. Five priority tiers determine the allocation of work (Table 4). Activation of each quantitative analysis depends on data coverage, independence and reliability. The decision precedes the results and is made independently of their direction.

Table 4. Priority tiers and activation rules

Tier	Content	Status and criterion
1. Core	M1, M2, M3a and the minimum M4 performance criterion	Confirmatory for Δ_{hist} within the adequate scope; the prespecified fallback expert criterion is activated if performance coverage is insufficient
2. Human validation	M3c (expectation), then M3b (categorisation), with the same listeners	A family of two confirmatory hypotheses (H_{pred}, H_{cat}), with Holm correction
3. Minimum M5	D_{fam} and D_{within} for Nikolaos of Smyrna	Predefined descriptive measures with work-level uncertainty
4. Supplementary studies	Broader M4; full M5 (D_{seq}, D_{func}); ATL; ATL-INV (R0–R3); R4; pilot M_{thesis} (P0–P2); A2	R0 starts in WP0; R3 and the ATL transfer test follow their own criteria; R4 begins after M2 is finalised; P2 follows the P1 identifiability criterion
5. Conditional extensions	Neural versions of ATL and M_{thesis}; synthetic training data; pretraining on a Gregorian chant corpus; joint estimation of boundaries and alignments; M3n (EEG/MEG); electroglottographic measures; a second repertory	Activated when the simpler model leaves measurable room for improvement, or when infrastructure, ethics approval and source access are confirmed

4.2 WP0: inventory, specification finalisation and adequacy criteria

Training rests on two validated forms of evidence: reliable family judgements and interpretable relationships between historical exegeses. The preparatory phase, WP0 (months 1–6), is the first decisive milestone, because the coverage it establishes determines the inferential status of the main comparison. WP0 compiles a registry by family and mode, recording works, occurrences, independent witnesses, related-source groups, verified exegetical and copying relationships, feasible shared-family pairs and available performance coverage. For each exegesis, the registry distinguishes autograph, manuscript copy and printed edition, recording attribution, editorial mediation and notational phase. Shelfmark, folio or page, and the precise musical passage identify each witness. Correspondences rendered ambiguously, partially or not at all in the exegesis are recorded accordingly.

To prevent circularity, specifications are developed in five steps: (1) drafting the candidate specification, stating the analytical level, diagnostic relations, contextual rules and exclusion criteria; (2) revision using development material only; (3) fixing the version; (4) independent reliability assessment by two experts on a new sample; and (5) construction of B2 and EISE training pairs solely from families that pass validation. A family with insufficient reliability remains a candidate or is removed from the main lexicon; its specification is not retrospectively adjusted in response to model results. Initial development cases are formulae associated with the *parakletike*, *kylisma* and *piasma*, drawing on Gheorghiță's comparative tables of older notation and exegesis in eighteenth-century Sunday Communion chants (2010, pp. 228–245). The *antikenoma*, *lygisma* and *heteron parakalesma* (Karagounis, 2003, p. 788) are examined as candidate hard negatives for Route A, subject to independent family judgement, and as candidate cases for historical recovery. Since many families bear the names of great hypostases, a single piece of evidence cannot fulfil both roles: family membership is judged from analytical New Method notation without access to the witness reading, and a family label derived from the very sign to be recovered is excluded from the inputs to historical recovery.

Two separate M2 criteria are first assessed at the end of WP0; the performance criterion is finalised after the measurement pilot and before M2 testing begins. The *pair adequacy criterion* (G_{pair}) assesses whether there is sufficient common support between verified parallel exegeses and matchable B2 pairs. The *performance-data adequacy criterion* (G_{perf}) assesses the overlap between historically documented formulae and recordings, with three outcomes: *adequate*, where coverage supports confirmatory analysis at the detailed internal coordinate; *borderline*, where it supports only a coarser, predefined regional division, at which confirmatory analysis is then conducted; and *inadequate*, where coverage or measurement reliability is insufficient. For the last case, a *fallback criterion* is declared in advance:

blind judgements by expert chanters outside the annotation team on which regions of each formula allow temporal freedom in performance, collected before M2 results are available. The resulting expert-based contrast, $\Delta_{\text{hist}}^{\text{expert}}$, is reported separately as a secondary measure of agreement with expert judgements. Δ_{hist} retains confirmatory status only over the scope supported by both criteria.

For each criterion, the deciding quantity, procedure for setting its threshold, finalisation point and excluded data are stated in advance (Table 5). Numerical thresholds are derived from power simulations using the coverage and dependencies measured in the inventory and explicitly assumed variances, and are fixed before the main comparisons.

Table 5. Adequacy criteria: decisions, finalisation points and excluded data

Criterion	Quantity	Threshold setting	Finalisation	Excluded data
G_{pair}	Verified parallel exegeses per family; families with matchable B2 pairs	Power simulation for Δ_{hist}	Month 6	Performance data, model results
G_{perf}	Families and regions with enough chanters; alignment reliability from double annotation	Precision simulation; minimum reliability per division; separate measurement pilot	Before M2 testing begins	Final performance set, model profiles
g_{min} (M1)	Family recall on development data	Simple reference model and prespecified expert judgements of useful recall and improvement	WP1, before grid training	Test set, historical hard-case set
Specifications	Agreement between two experts per family	Conventional reliability thresholds declared in advance	Before pair construction	M1–M2 results
G_{ATL}	Parallel exegeses and families available for holding out entire families	Minimum coverage for a transfer test	Month 6; reassessment at months 12–18	Transfer-test families
G_{rec0}	Readable witnesses, classes with multiple candidates, cost per passage	Criteria for initiating historical recovery	Month 6	Held-out test labels
G_{rec1}	Occurrences per label value, independent works and witnesses, reading agreement	Power simulation for $\Delta_{\text{rec}}^{\text{rule}}$	Months 12–18	Held-out test passages
P1 (M_{thesis})	Proportion of identifiable occurrences; agreement on family judgements for hard cases	Predefined minimum identifiability for P2	Month 14	Test works for Δ_{family}

4.3 M1: representation and formula segmentation

Once sources and annotation rules are fixed, the first computational question concerns the input. New Method information can be represented by graphic sign, atomic melodic event or composite neumatic unit. These forms differ substantially: the graphic sign preserves the written sign itself but sometimes combines several movements; the atomic event provides a common elementary reference scale but loses the scribe's grouping; and the composite unit preserves grouping through an explicit rule. The formula is the unit the model must *find* in these inputs. M1 examines which input best supports segmentation and family discrimination, and whether an explicit formula unit provides information beyond individual events.

The distinction between input unit and target unit is central. The input unit is how notation is read; the formula is what is sought in each reading. A kylisma, for example, may be read as a sequence of graphic signs, atomic movements or neumatic groups; in every case, the model must return the same continuous stretch of chant with its family label. The relationship resembles *named entity recognition* in language, where a model reads characters or subword units but returns labelled text spans. Here, the segment has variable and sometimes ambiguous length, and its label is defined musicologically.

Each source is converted into a **common symbolic reference layer**: a sequence of atomic melodic events linked bidirectionally to the original graphic signs. A cumulative notational coordinate aligns graphic signs, events, syllables, formulae and recording time points. On this layer, M1 compares twelve encodings in a crossed 3×4 design (Table 6).

In the grid, F denotes the feature set; elsewhere in the proposal it denotes the formula family, with the meaning clear from context. The twelve conditions are evaluated with a common boundary estimator, training protocol and metrics, and an equal tuning budget. The progressive addition of temporal, modal and textual layers is musicologically motivated, because the interpretative function of the qualitative signs depends on their combination with neighbouring elements, their position within the phrase and temporal organisation (S. K. Despotis, 2007); the experiment shows which layer is most discriminative. The comparison follows systematic evaluations of alternative encodings for

Table 6. The M1 $U \times F$ encoding grid

Factor	Conditions and information content
U**: input unit**	u_1: graphic sign as it appears in the source; u_2: atomic melodic event; $u_3 = f_{\text{group}}(N)$: composite neumatic unit with a deterministic, documented and reversible grouping rule
F**: features (cumulative)**	N, notational: relative movement and types of quantitative and qualitative signs; $+T$, temporal: temporal signs and derived relative values; $+M$, modal: current mode, martyria (pitch and mode indication), phthora (modal-change indication); $+L$, textual and prosodic: syllable, word, stress, word and phrase boundaries, melismatic density

machine learning under shared models and criteria, as in harmony (M. Kaliakatsos-Papakostas et al., 2025); here, the criteria are segmentation and family discrimination.

Because the representations yield different numbers of symbolic units, likelihood per unit changes both measurement scale and target, requiring common normalisation for a fair comparison. The information-theoretic comparison therefore defines a **common target** Y_t, the minimal melodic description of the next atomic event, and a common **context horizon** H_{ctx} of preceding events. Each representation R supplies only the context $C_R^{(H_{\text{ctx}})}(t)$, using completed units within the same atomic-event span, and cross-entropy is measured per atomic event:

$$\text{CE}_R = \frac{1}{N_{\text{evt}}} \sum_t -\log_2 P\big(Y_t \mid C_R^{(H_{\text{ctx}})}(t)\big).$$

All conditions thus predict the same target and differ only in the information available to them. The target and horizon are fixed in WP1. The causal restriction applies to the information-theoretic test; for segmentation, all encodings have the same contextual span.

The $U \times F$ grid compares input encodings. A separate experiment examines the **structural unit**: does the explicit formula itself provide information as a level of organisation? Each comparator corresponds to a different account of structural origins. An atomic-event Transformer (T_{evt}) learns long-range dependencies without formula labels; unsupervised Pitman–Yor segmentation induces recurrent segments solely from the corpus's statistical structure (Lanz and Hajič, 2025); the Local Boundary Detection Model (LBDM) proposes boundaries from local discontinuities in movement and timing (Cambouropoulos, 2001a); and segmentation through melodic parallelism places boundaries at the ends of repetitions (Cambouropoulos, 2006). Comparisons use the same targets and splits, showing whether formula boundaries are better explained by local articulation cues, repetition, statistical grouping or structured family knowledge.

The new task. To evaluate recognition in continuous works, M1 defines **joint formula segmentation and family recognition in continuous chant**. The estimator is a semi-Markov conditional random field (semi-CRF), which scores entire candidate segments of variable length. It is trained from partial annotations and keeps unannotated regions latent, avoiding their erroneous treatment as negative examples. Its likelihood incorporates more than one acceptable boundary pair $\mathcal{B}_x$ where musicological analysis supports alternative readings; it marks cases outside the available lexicon through an unknown-family category $\varnothing_F$, and provides calibrated abstention when evidence is insufficient. These are distinct decisions: $\varnothing_F$ is a taxonomic decision (the occurrence belongs to a family outside the known lexicon), while abstention is an uncertainty decision (the maximum family posterior falls below a development-set threshold for prespecified coverage). The decoder selects non-overlapping segments on the common coordinate at a prespecified analytical level. Nested formations and alternative levels of analysis are recorded separately in the annotation protocol. For partial annotation, the likelihood sums over complete analyses that contain the mandatory annotated occurrences with one of their acceptable boundary pairs; the precise constraint class and dynamic-programming state are fixed in the implementation protocol.

Selection rule. A representation may yield accurate boundaries yet insufficient family discrimination. The representation $R^\star$ is therefore selected by a predefined rule, exclusively on the fully audited development subset, split by work and related-source group. The primary measure $\widehat{M}(R)$ is pooled boundary F1 (micro-F1) under a common tolerance determined by independent annotation agreement; the family measure $\widehat{G}(R)$ is macro-averaged family recall on occurrences with externally documented boundaries. The rule is

$$\widehat{\mathcal{A}}_{\text{dec}} = \{R : \widehat{G}(R) \geq g_{\text{min}}\}, \qquad R^\star = \arg\max_{R \in \widehat{\mathcal{A}}_{\text{dec}}} \widehat{M}(R).$$

The threshold g_{min} is calculated before grid training as the larger of a minimum useful level and the performance of a simple reference model (nearest family medoid) plus a minimum useful improvement. The useful levels are

specified independently by two experts outside the evaluation team before they see any results. Selection is repeated over work-level splits of the development subset, and the stability of $R^\star$ is reported. If no representation passes the threshold, readiness for confirmatory M2 is recorded as unsuccessful and the threshold remains unchanged. Selection uncertainty and the loss in family performance relative to the best family-discriminating representation are reported separately.

After $R^\star$ is fixed, the **historical hard-case test** is conducted once: negative pairs with related metrophony and positive pairs with substantial analytical differences, starting from Karagounis's cases (2003, pp. 788, 810–812). AUROC measures how well the selected representation distinguishes them. The result is reported as an independent test and leaves the selection unchanged. The task itself is evaluated with metrics assessing boundaries and labels jointly: segment-level F1 with the correct family, where a segment is correct if its boundaries match an acceptable pair in $\mathcal{B}_x$ within the common tolerance; unknown-family detection AUROC, using different entire held-out families for calibration and final testing; and a risk–coverage curve for abstention. The structural-unit comparison uses cross-entropy per event (in bits) for models that provide probabilities (T_{evt}, Pitman–Yor), and the same boundary metrics for all comparators. LBDM and parallelism-based segmentation, which propose boundaries only, are assessed using the boundary metrics.

4.4 M2: the EISE encoder and the comparison of supervision sources

Representation selection determines how a formula reaches the model; M2 asks from which musical relation its shared identity is learned. Two occurrences of the same family in different works differ for many reasons: mode, scale degree, text and composer. Two exegeses of the same older passage share a documented textual and musical reference. Their differences locate expansions and contractions in the written renderings; source criticism examines their relation to exegetical practice, working sources and transmission. If this more specific relation better teaches which parts of a formula remain stable and which allow variation, a model trained on it should locate variation where living practice also places it. Because the question concerns the source of positive pairs, it has broader relevance to representation learning (§2.3).

Why a performance criterion. The criterion must remain outside training and provide an independent observation of within-formula variation. The study compares chanters performing the same specified written realisation. Differences in timing measure performed variability; their interpretation as flexibility within the tradition draws on the score, context and independent expert assessment. The working hypothesis is that regions of historical written variation and present-day timing variability coincide to a measurable degree. Using expressive timing as external evidence of musical structure has precedents in the relationship between boundaries and timing (Cambouropoulos, 2001a) and between expectation and timing (Gingras et al., 2016); M2 tests the specific historical–performance connection. Contemporary performance supplies a second, independent observation channel for musical structure, with its own historical and performing context. Two mechanisms predict opposite signs: training on parallel exegeses may organise local differences more effectively or compress them in seeking a shared identity. The estimand Δ_{hist} measures which outcome prevails, while separate diagnostics examine whether the results support either mechanism.

The shared encoder. EISE (Exegesis-Invariant Segmental Encoder, seeking a stable representation of identity across exegeses) is the shared architectural framework for all neural conditions; "EISE" also names the experimental condition trained on pairs of parallel exegeses. The encoder takes the fixed $R^\star$ and computes a local state h_t for each atomic event: an internal description of the event together with its context (Figure 4). For each candidate segment $[s, e]$, the states are pooled into a **family-identity representation** z_{id}, to which the M1 semi-Markov CRF scoring and a common weighted positive-pair attraction term are applied.

The relational term is attraction-only, distinct from a full contrastive objective: discrimination between families is supported by CRF scoring and a common family-prototype term. The main change across conditions therefore concerns the source and selection of positive relations. The historical hard negatives in §1.1 are reserved for independent evaluation. Local states are retained to study the internal form of realisations, and an auxiliary masked-event reconstruction objective, following symbolic music pretraining (M. Zeng et al., 2021), encourages preservation of local information. Since the relational term also affects the shared encoder, the adequacy of this protection is tested through separate measurements on the identity branch and local states.

Four comparison conditions are evaluated on the same eligible data (Table 7). The contrasts distinguish the contributions of the selected components: B0 versus B1 assesses representation learning; B1 versus B2 assesses the relational term with family pairs; and B2 versus EISE assesses the *provenance* of positive pairs, the object of the main comparison. The estimand Δ_{learn} summarises the total gain of the full condition over the symbolic description.

Operational definition of the criterion. For each family F, all occurrences are aligned to a common internal coordinate, divided into K predefined regions (onset, internal regions, cadence), with K fixed in WP1. Alignment is

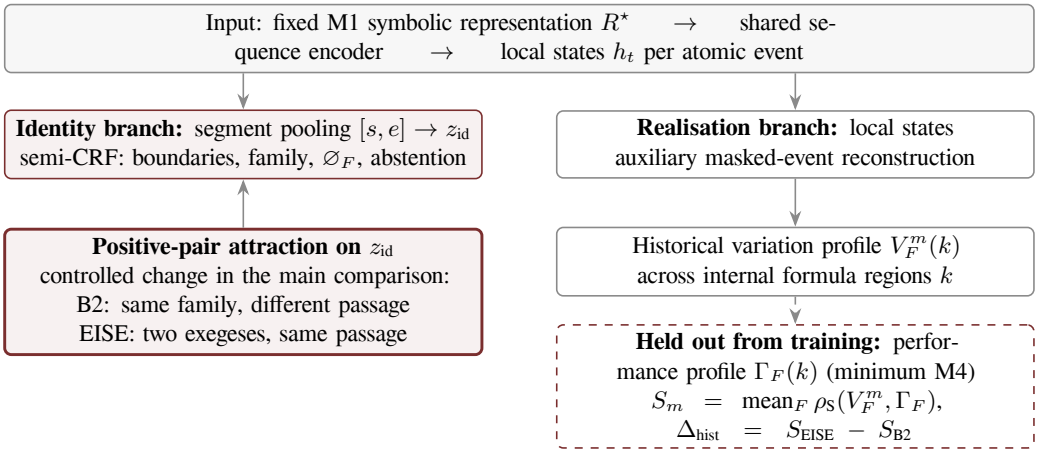


Figure 4. Shared M2 architecture. B2 and EISE vary the provenance and selection of positive relations, subject to the matching controls described in the text; B1 uses the same encoder without a pair term, and B0 applies the same profiling procedure to symbolic features. The performance profile is used exclusively for evaluation.

Table 7. The four comparable M2 conditions

Condition	Supervision and estimation	Question addressed
B0	Strong direct symbolic description of historical variation, without representation learning	How much of the external profile is already explained by the best simple symbolic description?
B1	Same encoder, partial segmentation and family supervision, without a relational term	What do the shared neural infrastructure and family label provide on their own?
B2	Same encoder, with positive pairs from the same family in different passages	What does relational supervision from a shared analytical category provide?
EISE	Same encoder, with positive pairs from verified parallel exegeses of the same passage	What does relational supervision from a shared historical object provide?

computed once on the symbolic input, using the specification's boundaries, before any training, so that all conditions share the same comparison coordinate. Since alignment errors may affect representations differently, their effects are tested through common perturbations and a blind reliability assessment. For model m, variation in region k is the within-family dispersion of the mean local states $\bar{h}^m_{a,k}$ of occurrences a belonging to F:

$$V_F^m(k) = \tfrac{1}{2}\ \text{mean}_{a \neq b} \left\| \bar{h}^m_{a,k} - \bar{h}^m_{b,k} \right\|^2.$$

The profile V_F^m is computed over all occurrences of F in the common evaluation material and measures the learned within-family organisation of variation; B0 follows the same procedure with symbolic features in place of h.

The performance profile $\Gamma_F(k)$ is the between-chanter variance of normalised log timing in the same region, accounting for overall tempo, local event density and the chanter's teaching lineage; pitch is a secondary variable. Principal teachers, training history and self-described performing tradition are recorded separately. Pedagogical groups used in analysis are defined before testing; where coverage permits, Γ_F is also computed within these groups to assess the influence of training background. Divergence between exegetes is the mechanism through which EISE training is expected to shape the organisation of variation. As a descriptive check of this mechanism, the exegetes' divergence profile on the matched passages themselves, and its correlation with Γ_F, are reported separately.

Agreement is $S_m = \text{mean}_F\ \rho_S(V_F^m, \Gamma_F)$. Rank correlation compares the ordering of regions without requiring the profiles to share a numerical scale. This property applies to monotonic transformations of profile values, whereas geometric distortion of a space can alter rankings; anisotropy and dimensional collapse are therefore assessed through prespecified diagnostics, and families with undefined correlations are handled by a common rule. Since within-family rank correlation is coarse when there are few regions, sensitivity analyses use Kendall τ_b and a hierarchical region-level model. Uncertainty is estimated through cluster bootstrap resampling at work and source-group levels. Once training is finalised, performance results are excluded from tuning. The comparison is repeated on progressively larger subsets of documented pairs (supervision curves) to determine how much historical supervision is required and when gains saturate.

Eligibility and matching. An EISE positive pair consists of two separately documented analytical renderings of the same passage, provided their exegetical relationship, passage homology and, independently, shared family membership have been verified. Copying relationships or a shared working source are recorded from provenance evidence. Shared family membership is judged independently of the same-passage link: pairs in which exegesis changes the

family attribution are excluded from supervision and studied separately. The design matches each occurrence's exposure, the pairs' surface difficulty and the number of positive relations. Since historical edge selection may alter the supervision graph's structure, family coverage, node degrees, occurrence reuse and distance distributions are recorded and matched where feasible. Matching quality is documented quantitatively, residual differences are examined through prespecified sensitivity analyses, and the comparison concerns the common eligible support provided by actual pairs.

Controlling the textual layer. Two parallel exegeses of the same passage necessarily share the same poetic text and a similar syllabic layout. The encoder could therefore use text as a shortcut independently of the melodic relation. If $R^\star$ includes layer L, M2 includes a sensitivity analysis that removes it before the shared encoder processes the input ($R^\star_{\setminus L}$), with derived text-dependent features recomputed accordingly, and, where feasible, a comparison using pairs that share the same textual passage in different compositions.

A2: when diagnostic musicological knowledge helps. The specifications describe relations whose assessability varies by witness, and allow several sufficient configurations of the same family. If their use improves learning, the gain may arise from their musicological content or simply from the structure added to the model. A2 separates these causes through a 2×2 factorial design crossing *valid* or *systematically perturbed* knowledge with *flat* or *structured* use; the interaction shows whether structured use is beneficial specifically when the relations are musicologically valid. Perturbations preserve prespecified frequencies and relation assessability, while observed absence, unknown status and non-assessability retain separate values. The operational forms of flat and structured use are specified in the WP1 technical protocol before any training, with common capacity and budget across all four conditions. Reported measures include segmentation, family recall, coverage of rare configurations and retention of information about the specific realisation. A2 is an M2 subanalysis within RQ2: it uses the same specifications, data and splits, varies only the two factors, and is reported separately from Δ_{hist}, which remains the primary M2 estimand.

4.5 M3: musicological and cognitive validation

M2 measures how the source of supervision affects the representation. A different question remains: do the computational categories correspond to relations recognised by independent experts and listeners? M3 separates expert analytical verification of the lexicon (M3a) from the use of categories in listening (M3c and M3b), keeping analytical validity and cognitive function as distinct outcomes.

M3a: independent experts. Experts in Byzantine musicology, selected through predefined experience criteria, apply the fixed specifications to new material under blinded conditions. The reliability sample must include an annotator outside the principal editor's teaching tradition. Blinding covers source identity, historical pairing and model results; experts retain the musical context required by the specification. They judge boundaries, satisfaction of each active diagnostic feature and final membership (member, non-member, unresolved) in sequence. Disagreements are classified by cause (boundary, analytical level, modal issue, diagnostic feature, insufficient source, historical name). They document the reproducibility of the specifications and competing readings; their musicological significance is assessed against the sources, without feeding back into M1–M2.

M3c: melodic expectation. The first listening session examines whether formula families contribute to predicting melodic continuation. A continuation may be locally predictable at pitch level yet unexpected in its family sequence, or vice versa. Listeners hear an excerpt and its continuation and judge how unexpected the continuation is. A variable-context probabilistic model (Prediction by Partial Matching, following the cognitive framework of IDyOM; Pearce, 2018) computes I_F for the next family exclusively from preceding context, while a corresponding event-level chain computes I_{event}. The stimulus bank balances both off-diagonal combinations (high I_F with low I_{event}, and vice versa), allowing H_{pred} to be tested without confounding the two levels.

M3b: categorical organisation. The second session, with the same listeners, examines whether shared family organisation is perceived across surface differences. Listeners hear three renderings and choose the one that fits least well with the other two. Mixed triads are selected from a common range of surface distances, so that family membership is distinguishable from simple acoustic similarity. The task applies the distinction between categorisation by diagnostic features and overall surface similarity to formulae (cf. Cambouropoulos, 2001b; Volk and van Kranenburg, 2012). The hypothesis H_{cat} is tested with a mixed-effects triadic-choice model, using crossed random effects for listener, stimulus, triad and family. Secondarily, perceptual-space geometry is derived from triadic judgements without family labels and compared with B2 and EISE representations through within-family representational dissimilarity matrices (RDMs). Generalisation of family discrimination across conditions is assessed through cross-condition generalisation performance (CCGP), using separate fitting and test stimuli. Running M3c before M3b with the same listeners protects expectation judgements from exposure to explicit family comparisons. Participants

include experienced chanters and musicians without Byzantine training, and the stimulus bank is fixed after the M3a lexicon.

M3n: neural extension. Using the same stimuli and predictors, M3n would examine with EEG/MEG whether I_F adds neural explanatory power beyond I_{event} and acoustic features (cf. Di Liberto et al., 2020; Kern et al., 2022). It belongs to priority tier 5 and is activated only with written confirmation of infrastructure, expertise and separate ethics approval; the main hypotheses can be tested independently through behavioural measurements.

4.6 M4: contemporary performance variability

M4 measures how contemporary chanters vary timing and pitch in the same written passages. The **minimum component** (Tier 1) delivers the performance profile by month 36, after model finalisation and before calculation of Δ_{hist}. The **broader component** (Tier 4) analyses temporal and pitch variation at the resolution supported by coverage. The relationship between contemporary performance and historical written variation remains the explicit M2 hypothesis.

The minimum criterion is planned around targeted recordings of common written passages selected for their historical and family coverage; eligible DAMASKINOS recordings can supplement this material. WP0 establishes the recording plan, recruitment requirements and access arrangements. Recordings are aligned with the written source on the common notational coordinate. Temporal analysis distinguishes overall tempo, normalised local timing in each internal formula region, cadential lengthening and event density; each chanter's teaching lineage is recorded. Where synchronised electroglottographic data are available, measures of vocal-fold contact are examined exploratorily (priority tier 5). The measurement pilot determines the number of chanters and passages required within the available recording and annotation budget. Passage selection precedes M2 evaluation, is determined by musicological coverage and measurement reliability, and depends on consent, rights and independent adequacy assessment.

4.7 M5: from formula to complete work

The preceding studies examine individual formulae, their family relationships and their perception in listening and performance. M5 extends the analysis to complete chants, as anticipated in Figure 1, and applies the validated framework to the nineteenth-century compositions of Nikolaos Protopsaltes of Smyrna (I. G. Toma, 2018; Ch. Ioannidis, 2022). This choice serves three purposes. Nikolaos composes within the New Method in genres whose earlier tradition had already been rendered into analytical notation through exegesis, principally by Gregorios and Chourmouzios; his works show how the formula lexicon is used when a composer writes directly in the new notation. His relationship to earlier tradition remains an open question, and work-by-work comparison struggles to distinguish whether he *selects* different formulae or *realises* the same formulae differently. Finally, Ioannidis's published analysis (2022) of the Communion chant *Eis pasan ten gen* ("Into all the earth"), in relation to Petros Lampadarios and Chourmouzios, offers an independent point of comparison.

Each composition is characterised against a predefined, chronologically controlled reference corpus Ω along four dimensions, reported separately. Comparisons are stratified by genre and mode, with textual scope and passage length recorded to assess comparability. Each dimension is expressed as the work's percentile in the distribution of composers in Ω, so that the central question—whether Nikolaos departs more in selection (D_{fam}) or realisation (D_{within})—is assessed on a common scale through a prespecified comparison. *Selection* (D_{fam}) concerns which families the composer uses relative to the distribution in Ω. *Internal realisation* (D_{within}) concerns how he varies realisations within the same family; it is measured in the local states of the fixed EISE and, as a learning-independent control, in the B0 symbolic description. *Sequencing* (D_{seq}) concerns how he orders and recombines families into continuous chains, and *functional role* (D_{func}) concerns the structural, modal and textual contexts in which he places each family.

The **minimum M5** (priority tier 3) produces a work map, coverage rate and the first two dimensions with work-level uncertainty. The **full M5** (priority tier 4) adds sequencing and functional role where continuous transitions and independent functional evidence are sufficient; a composer-level reference distribution additionally requires a minimum number of independent composers, set in WP0. The map keeps connective, unknown and unresolved regions visible; each such region interrupts transition calculations to prevent artificial sequences across gaps in documentation. Where modal information has already been used in family attribution, functional role is assessed through independent evidence to avoid circularity. If independent musicological analysis supplies several acceptable maps of a work, indices are calculated for each and reported as sensitivity ranges, separately from statistical intervals.

Nikolaos's works are withheld from training, model selection and specification development; if a specification requires revision using his material, the relevant work is withdrawn from the application set. Ioannidis's analysis defines a prespecified comparative question: agreement or disagreement between the computational description and that analysis is tested before extending to further works. Each quantitative value is linked bidirectionally to

the passages from which it was derived. Statistical differentiation denotes departure from the declared Ω and is interpreted in relation to the particular passages and the historical conditions of the reference corpus.

4.8 Historical route: transformation, recoverability and realisation

The historical source that connects pairs in M2 can itself become a prediction target. Consider an older passage, the witness recording it and its analytical renderings. Three questions arise. How does Gregorios's rendering differ from Chourmouzios's for the same passage, and do the same kinds of change recur in formulae from other families? If the witness writes a petaste in one passage and an oligon in a metrophonically equivalent passage, or records a different great hypostasis over related metrophony, can the exegesis alone predict which sign was written? Setting older notation aside, how predictable is a formula's particular analytical form when its family and context are known? The example is schematic; the classes studied are selected through direct examination of witnesses.

These questions correspond to three tasks with different target variables (Table 8): ATL predicts the *other exegesis* X_b, ATL-INV the *older label* H in a specific witness, and the pilot M_{thesis} the *realisation* X given a known family. They are trained and evaluated separately, using shared alignment infrastructure; a common probabilistic language keeps their targets and conclusions distinct.

Table 8. Three complementary aspects of musical variation and the models used to study them

	ATL (transformation)	**ATL-INV (recoverability)**	M_{thesis} **(realisation)**
Musical question	How does one attested realisation change into another? Do regularities transfer?	What remains readable of older notation in the exegesis?	How predictable is a realisation when its family is known?
Direction	One exegesis $\rightarrow$ the other, $q_\phi(X_b \mid X_a, d_F, Z)$	Exegesis and context $\rightarrow$ older label, $q(H \mid M, Z, X_G)$	Family and context $\rightarrow$ realisation, $q_\theta(X \mid F, C)$
Primary estimand	Δ_{ATL} on held-out families	$\Delta_{\text{rec}}^{\text{rule}}$; also Δ_{rec}, Δ_{2ex}	Δ_{family}
Comparator	Fixed theoretical rule; simple edit-distance transducer	Metrophony and context; known exegetical rules; simple empirical mapping	Same model without family, $q(X \mid C)$
Test set	Entire families withheld from estimation	Held-out works with blind witness readings	Held-out works and source groups
What is learned	Which expansions and contractions are regularities of exegetical practice, and which are specific to a family or exegete	Which older distinctions can be recovered, under which documented conditions, and how far recovery exceeds known rules	How strongly family constrains realisation, and where the tradition allows flexibility
Status	Tier 4; criterion G_{ATL}	Tier 4; R3 subject to G_{rec1}	Tier 4, pilot (P0–P2)

ATL: Attested Transformation Learning. In a verified pair of exegeses of the same passage, alignment identifies expansions, contractions and other differences between the analytical renderings. ATL models these correspondences in a declared prediction direction; historical dependence is established separately from the sources. The musical question is whether the changes follow regularities of exegetical practice that extend beyond a single family. If a form of contraction, for example in a cadential movement, recurs across families under similar contexts, a model trained on some families should predict the test families more accurately. The test measures transfer between families for prespecified directions between exegetes. Limited transfer is interpreted alongside coverage, alignment reliability and model adequacy.

The family F identifies which occurrences are compared, and modal, prosodic and historical context Z defines the permissible conditions of transformation. In the new-family test, the input is a prespecified family description, d_F, expressed through shared diagnostic features. The encoding scheme is fixed before testing. The description is documented from material independent of the target rendering X_b, and entire test families are excluded from training and tuning. The prediction $q_\phi(X_b \mid X_a, d_F, Z)$ is tested only where the description is legitimately available. A probabilistic transducer with variable-length alignments learns correspondences from actual pairs, starting from musical edit distance with fragmentation and consolidation operations and stochastic string edit distance (Mongeau and Sankoff, 1990; Giraud et al., 2014; Ristad and Yianilos, 1998). A neural version with monotonic attention (S. Wu and R. Cotterell, 2019) is considered only at priority tier 5. Musicological mapping separately records a change in an active diagnostic relation, a transition to another permissible configuration and any change in family membership.

The transfer test withholds entire families from estimation and measures, in bits per atomic event, the improvement in predicting one rendering from the other:

$$\Delta_{\text{ATL}} = \text{CE}_{q_{\text{ref}}}\big(X_b \mid X_a, d_F, Z\big) - \text{CE}_{q_\phi}\big(X_b \mid X_a, d_F, Z\big),$$

where q_{ref} is the stronger, on development data, of a fixed theoretical rule and a simple empirical transducer. Cross-entropy is the total negative $\log_2$ probability of target renderings divided by their total number of atomic events, with common treatment of sequence termination; all comparators score the same target renderings. Distance between the prediction and actual rendering is reported secondarily. The alignment protocol and eligible directions are fixed before testing. If actual pairs are insufficient for the test, ATL delivers a descriptive morphology of transformations.

ATL-INV: from exegesis to older notation. ATL compares two analytical forms of the same passage. ATL-INV follows the older-to-analytical notation channel in reverse: it starts from the analytical rendering and asks which historically eligible notation the older witness recorded. The question connects research on the great hypostases (Alexandru, 2012) and Karas's position on qualitative distinctions (§2.1) with a testable inverse problem. Shared metrophony among candidates makes the problem controlled. The reference model $q(H \mid M, Z)$ uses metrophony and permissible context; adding the exegesis allows the predictive value of analytical notation to be measured. Expansions, temporal and qualitative signs, and syllabic distribution are candidate features of this information. A stable gain on independent works establishes a usable trace of the older distinction. A small or indistinguishable gain is assessed alongside uncertainty, calibration and model adequacy. The target is the witness's recorded written choice; historical oral practice constitutes a separate level of musicological evidence.

Let H be the older notational label, read blindly in a specific witness; M the metrophonic class with candidate set $\mathcal{H}(M)$; Z the permissible historical and modal context, containing no information derived from H (including any family label derived from the same older sign); and X_G, X_C the exegeses by Gregorios and Chourmouzios, where verified. The *cross-entropy* of a probabilistic model q is the mean negative base-2 log probability it assigns to the label actually read, $\text{CE}_q = -\frac{1}{N}\sum_i \log_2 q(h_i \mid \cdot)$. A lower value denotes better probabilistic prediction, and the difference between two values on the same passages measures, in bits per passage, how much additional information one model exploits. All models assign probability over the same $\mathcal{H}(M)$, refined by the same historical constraints, and are evaluated on the same held-out works; N counts passages, each with one label.

The strong comparator is defined before the primary estimand, because it determines what "recovery" means. Known exegetical rules, recorded before examination of the pairs (§2.1), become a probabilistic comparator, keeping evaluation well-defined under ambiguity or erroneous exclusions. Let $\mathcal{R}(X_G) \subseteq \mathcal{H}(M)$ be the labels compatible with the rules, and $q_0(H \mid M, Z)$ a smoothed prior with positive probability for every eligible label. Then

$$q_{\text{rule}}(h \mid M, Z, X_G) = (1-\varepsilon)\,\frac{q_0(h \mid M, Z)\,\mathbf{1}[h \in \mathcal{R}(X_G)]}{\sum_{h' \in \mathcal{R}(X_G)} q_0(h' \mid M, Z)} + \frac{\varepsilon}{|\mathcal{H}(M)|},$$

with $q_{\text{rule}} = q_0$ when $\mathcal{R}(X_G)$ is empty, and a smoothing parameter $0 < \varepsilon < 1$ selected on development data. The second comparator, q_{emp}, estimates smoothed label frequencies from a simple description $\phi(X_G)$ in training, under the same constraints (M, Z). The model $q_{\text{ref,INV}}$ is the stronger of the two on development data, selected by a prespecified rule. Three contrasts are then defined:

$$\Delta_{\text{rec}} = \text{CE}(H \mid M, Z) - \text{CE}(H \mid M, Z, X_G), \tag{1}$$

$$\Delta_{\text{2ex}} = \text{CE}(H \mid M, Z, X_G) - \text{CE}(H \mid M, Z, X_G, X_C), \tag{2}$$

$$\Delta_{\text{rec}}^{\text{rule}} = \text{CE}_{q_{\text{ref,INV}}}(H \mid M, Z, X_G) - \text{CE}_{q_{\text{inv}}}(H \mid M, Z, X_G). \tag{3}$$

The estimand $\Delta_{\text{rec}}^{\text{rule}}$ is **primary**: it shows whether the recovery model q_{inv} extracts information about older notation from the exegesis beyond the strongest comparator. The two individual comparisons are also reported, distinguishing gains beyond published rules from gains beyond simple empirical mapping. The estimand Δ_{rec} gives the total gain from the first exegesis over metrophony and context. The estimand Δ_{2ex} is calculated exclusively on the same eligible passages with two exegeses and measures the second exegesis's added value. Dependence between exegetes (shared source, copying) is recorded in the registry and represented through a directed probabilistic model, with the reverse direction as a sensitivity analysis. Differences are measured on held-out works against a prespecified smallest effect size of interest. The population-level non-increase of optimal conditional entropy concerns unrestricted optimal predictors; estimated differences between the specific models may have either sign.

The generative formulation is straightforward. The forward channel describes how an older written choice H, within permissible context (M, Z), is rendered in analytical notation X_G. If the probability of each analytical

rendering under each older choice is known, Bayes' rule gives the probability of each older choice after observing the exegesis:

$$p(H \mid X_G, M, Z) \;\propto\; p(X_G \mid H, M, Z)\, p(H \mid M, Z).$$

The channel is estimated on ATL-INV training passages, with choices made solely on development data, and compared with direct discriminative prediction of H where material is sufficient. The generative formulation yields interpretable profiles of analytical features associated with each older label, compared within the same metrophonic class and common context. These are profiles of exegesis where the source relationship is established, and profiles of correspondence to the recorded witness where the working source remains uncertain.

Non-trivial recovery and the scope of the target. Passages in which a sign survives unchanged in the New Method are recorded separately as controls; the main set requires recovery through the exegesis's melodic organisation. The label H concerns the recorded witness, whereas the exegete's working source is usually unknown. For passages with several older witnesses, their agreement on H is measured as an empirical indicator of the historical target's stability; this alone does not determine achievable predictive performance. Disagreements are reported as findings about the tradition. Unresolved distinctions are accompanied by a record of observable conditions: agreement or disagreement between witnesses, correspondence uncertainty, known details of the working source, coverage and model error. Attributing a discrepancy to historical change requires additional evidence.

Stages and inferential status. Historical recovery proceeds through four stages, each with a different prerequisite. **R0** compiles the registry and blind palaeographic readings, initially targeting 10–20 actual passages. Each witness is read separately by two readers, including at least one external palaeographer, without access to the exegeses. Agreement is measured through a prespecified index and acceptance threshold, and disagreements are resolved through a documented procedure. The adjudicated readings provide the reference labels (ground truth) for computational evaluation. The initial sample tests feasibility; the held-out test set is formed after expansion assessed by G_{rec1}. **R1** uses the grammar of older notation to define historically eligible classes of older labels with multiple candidates. **R2** numerically verifies the estimators on controlled synthetic examples with known labels, validating the computation. **R3** applies recovery to actual independent passages, starting with individual signs and progressing, where material permits, to relations between signs and complete complexes.

The month-6 criterion (G_{rec0}) concerns access, readability and correspondence; the months 12–18 criterion (G_{rec1}) concerns class coverage, work independence and reading reliability. If the latter is met, $\Delta_{\text{rec}}^{\text{rule}}$ is evaluated confirmatorily within the scope supported by the inventory; otherwise, R3 is delivered as a documented case study. The inventory, readings and documented ambiguities are delivered in either case to the extent permitted by the sources; a quantitative recoverability map and calibrated predictions require an adequate independent evaluation set.

R4: access to historical information in the representations. After the main M2 comparison is completed, the B1, B2 and EISE encoders are fixed, with parameters held constant throughout R4. Simple capacity-limited probing classifiers test whether the older label H is predictively accessible from the finalised representations, using common historical splits, budgets and control tasks (Hewitt and Liang, 2019). The primary measure is probe cross-entropy on held-out passages at fixed capacity; description length is secondary, indicating how efficiently the label can be learned from each space (Voita and Titov, 2020).

The main comparators are B1, which uses the same shared encoder without a relational term, and the symbolic input, both in full form (u_2, the atomic melodic event, with all feature layers $N{+}T{+}M{+}L$) and as the selected $R^{\star}$. B2 is reported separately because it is trained to bring different passages of the same family closer, thereby affecting passage-specific information such as H in its own way. Test passages and all their parallel exegeses are excluded from training the compared encoders. R4 measures the *accessibility* of historical information under stated capacity and is reported separately from Δ_{hist}. The interpretation of each outcome is specified in advance: EISE outperforming or underperforming B1 denotes a corresponding difference in predictive accessibility at that capacity; an indistinguishable difference is interpreted through its uncertainty interval and measurement reliability, while equivalence is assessed only within prespecified bounds. An interpretation in terms of compression additionally draws on the separate diagnostics.

M_{thesis}: from family to possible realisation. Historical recovery concerns information about older notational choices. A related but independent question concerns melodic realisation itself: when a formula's family and local, modal and prosodic context are known, which specific realisations are probable? This connects recognition of a musical structure with its generative variability, the relationship between type and realisation at the heart of composition through formulae. The pilot study M_{thesis} turns this into a measurable quantity. The name uses the transliterated Greek term *thesis*, "melodic formula". M_{thesis} is a conditional probabilistic model of realisation: it connects a known family and context with its possible melodic forms. The type–realisation relationship provides a basis for future transfer tests in other traditions with melodic types.

The model is defined as a *forward* conditional model $q_\theta(X \mid F, C)$: X is the realisation in the target region, F the family and C the local, modal and prosodic context, containing no information derived from X; the target is hidden before context encoding. The reference model $q(X \mid C)$ shares the same target, training, budget and held-out works, and the estimand

$$\Delta_{\text{family}} = \text{CE}(X \mid C) - \text{CE}(X \mid F, C)$$

is measured in bits per atomic event of the target region, with a common target and normalisation, and assessed against a prespecified smallest effect size of interest. A positive value denotes additional predictive value of the known family in the particular model; a small or indistinguishable value defines the observed gain with its uncertainty. Since a family is partly described through relations within the realisation itself, an intermediate model $q(X \mid C, \sigma_F)$ uses only the specification's required constraints. The identity

$$\Delta_{\text{family}} = \big[\text{CE}(X \mid C) - \text{CE}(X \mid C, \sigma_F)\big] + \big[\text{CE}(X \mid C, \sigma_F) - \text{CE}(X \mid F, C)\big]$$

separates the predictive value of those constraints from the additional value of the full family label. This algebraic decomposition allows the predictive contributions of required constraints and the full family label to be measured separately. The analysis also checks whether the required constraints already identify the family uniquely; in that case, the second term mainly concerns estimation and modelling differences.

The provenance of the family label is recorded separately because it determines the type of inference. If the annotator has seen the target realisation, the result describes predictability given a known label. Prospective prediction is evaluated only on the subset whose family is documented from information available outside the target region.

The model also supplies a measure of *typicality*: $-\log_2 q_\theta(x \mid F, C)$, normalised per atomic event to compare realisations of different lengths, indicates how expected a realisation is within its family and context relative to a declared reference corpus. Typicality is distinct from musicological *acceptability*, judged through the specification and independent experts: a rare but acceptable realisation is recorded with its uncertainty and the reasons for its family membership, while ambiguous cases follow a separate assessment protocol. These two axes connect probabilistic modelling with musicological judgement: typicality captures a realisation's predictability given family and context, while acceptability documents compatibility with the specification and tradition. If the pilot shows that family measurably constrains realisation, the same framework—from type and context to permissible realisations—can be tested in other modal traditions with melodic types (§9), subject to the same identifiability checks.

The study has three stages. **P0** (by month 8) delivers a registry of observable diagnostic relations, known and unknown features, and ambiguous occurrences. **P1** (by month 14) examines identifiability through hard examples, independent family judgements and justified abstention, and determines the next phase's scope. **P2** (by month 22) compares simple probabilistic models, such as variable-order models and nearest-example models using edit distance, and estimates Δ_{family} and typicality within the eligible scope if material is sufficient. P0 and P1 deliver the registry and identifiability report in either case. A neural decoder, for example a recurrent encoder–decoder (K. Cho et al., 2014), is considered at priority tier 5 only when independent observations are sufficient and measurable room for improvement remains over simple reference models. After M2 is finalised, the fixed B1, B2 and EISE representations may be used as context encoders, with the target hidden before encoding.

Two inversions with different targets. Inverting M_{thesis}, $p(F \mid X, C) \propto q_\theta(X \mid F, C)\, p(F \mid C)$, yields probabilistic family recognition from an observed realisation. ATL-INV follows a different historical channel, from an older notational choice to its analytical exegesis, and predicts a different variable, H. A generative comparison of M_{thesis} family recognition with M1 remains exploratory and has its own protocol.

Conditional extensions. If the inventory shows that uncertainty in boundaries and historical alignment systematically affects the analyses, joint estimation of boundaries, families and alignments is considered, with a strong sequential baseline and its own protocol; at final testing the model receives a single available musical input. Neural versions of ATL and M_{thesis}, synthetic variants for training and pretraining on a Gregorian chant corpus belong to priority tier 5. Synthetic items are excluded from historical evidence and evaluation sets.

4.9 Evaluation principles, preregistration and reproducibility

Five principles apply across M1–M5 and the historical and pilot studies.

- **Data splits.** Training, development and held-out test sets are defined by complete work and related-source group before pair construction; segments of the same Cherubic Hymn setting or exegetical chain cannot appear in both training and evaluation.
- **Development-set decisions.** The representation $R^\star$, architecture, hyperparameters and reference models are selected exclusively on development data.

- **Preregistration.** Primary estimands, statistical analysis rules (cluster bootstrap resampling by work and source, Holm correction for the declared family of two human-study hypotheses) and adequacy criteria are registered before access to the main evaluation data.
- **Strong simple reference models.** Neural models are compared with strong non-neural or simpler baselines (B0, specification and exegetical rules, event models) and examined through ablations, so that performance can be attributed to the musical information under study (Sturm, 2014).
- **Open science.** Code, the provenance registry, specification versions and the complete technical protocol are made openly available; for material with redistribution restrictions, metadata, provenance identifiers and reproduction instructions are published.

The two routes share the provenance registry and splitting rules, while each task has its own targets, held-out test sets and inferential status. Related sources are also kept together across tasks, so that fitting material for one task cannot become final evaluation material for another. Historical labels are used for R4 only after the corresponding encoders and M2 protocol are fixed.

5. Data, resources, feasibility and ethics

The methodology uses several forms of evidence, each with a distinct function: catalogues suggest candidate families; complete works provide continuous sequential context; parallel exegeses document same-passage relationships; older witnesses supply historical labels; and recordings provide external performance observations. Their actual availability and overlap are assessed in the WP0 inventory, which also measures the cost of converting them into auditable data.

5.1 Corpora

Table 9 shows the role and current status of each material category. Its actual coverage determines which comparisons can be supported by reliable data.

Table 9. Corpora, roles in the research design and current status

Corpus	Content	Role	Status
***Papadic Formulae*, St Anthony's Monastery**	Papadic melodic types drawn from 758 compositions (248 Greek Cherubic Hymn settings, 101 non-Greek Cherubic Hymn settings, 351 Communion chants, 58 other works), with work and composer recorded for each type	Catalogue starting point, provenance documentation, candidate family relations for validation	Publicly available catalogue
Complete New Method Cherubic Hymn settings and Communion chants	Complete papadic works in the common symbolic layer, with traceable provenance	M1–M2 training and evaluation; M5 reference corpus Ω	To be digitally encoded; scope determined in WP0
Parallel exegeses by Gregorios and Chourmouzios	Pairs of analytical renderings of the same passage, with documented attribution; autographs, manuscript copies and printed editions recorded separately	EISE positive pairs; ATL data; ATL-INV input	To be inventoried and verified in WP0
Older witnesses	Targeted reading of passages with an exegesis, blind to that exegesis	Historical labels for ATL-INV and R4	Collection target; pilot of 10–20 passages
DAMASKINOS corpus (D. Delviniotis and G. Kouroupetroglou, 2011)	Recordings of 20 chanters performing a common ecclesiastical repertory under standardised conditions	Candidate source for M2's performance criterion and broader M4	Existing corpus; access and overlap assessed in WP0
Supplementary and targeted recordings	Multiple chanters performing common papadic passages	Planned route to the minimum performance criterion; scope and recruitment fixed in WP0	Collection target, subject to consent and rights

The 758 compositions describe the catalogue's source pool; the inventory determines the final number of eligible examples. Language, date, genre, editorial mediation and the catalogue's selection policy are recorded separately. Each retained occurrence is checked against its source edition or manuscript, including its exact location; source strata and duplicate transmission are tracked throughout evaluation. Since the catalogue strongly represents cadential formations, coverage is recorded separately by functional role (opening, connective, internal and cadential formulae), so that each conclusion is accompanied by coverage of the relevant role. The published DAMASKINOS

inventory includes Cherubic Hymn settings; overlap with the planned Communion-chant sample requires verification. WP0 measures the exact overlap between historical formulae and recordings and the distribution of chanters by training background, using the semi-automatic Praat annotation methodology of G. Chryssochoidis, D. Delviniotis and G. Kouroupetroglou (2007) where available.

Historical pilot corpus. The initial historical sample (10–20 passages, stage R0 in §4.8) connects an older witness, one or two exegeses and manual alignment. Blind readings are recorded separately from assisted readings, and the registry distinguishes the recorded witness from the exegete's possible working source. Pilot digital encoding may use appropriate adaptations of the MEI neume schema after schema validation and round-trip conversion checks, since encoding Middle Byzantine notation remains under development.

5.2 Infrastructure, tools and access

The technical infrastructure maintains bidirectional correspondence between the primary notational document and the symbolic unit entering the model, allowing every computational result to be checked against its source passage. Structured files from the open-source notation software Neanes (Neanes Project, 2026) are imported through deterministic adapters that preserve graphic signs, modal and temporal indications, text and provenance. Optical music recognition for printed editions is an optional aid, subject to measured conversion error and full manual verification. The research repository `byzantine-chant-representation` (version 0.1.0) already implements conversion of Neanes files into canonical melodic-event sequences and the twelve representations of the $U \times F$ grid; WP1 will audit it using a recorded version, reconstruction checks and representative actual files. This documents the project's technical preparation; corpus readiness is assessed in WP0. WP1 develops the version-controlled family registry and experiment-reproducibility repository. Since the symbolic corpora are small to medium in size, models are small and trained on a single mid-range graphics accelerator, with a common computational budget across compared conditions. Listening tasks and acoustic analyses are designed for standard recording and listening-test equipment; WP0 documents equipment access, personnel commitments and annotation costs. WP0 records usage rights for every digitisation, edition and recording.

5.3 Researcher role and safeguards for impartiality

The candidate is an active chanter with extensive practical experience, training in the teaching lineage of Simon Karas and Lykourgos Angelopoulos, and a diploma in Byzantine music, as well as a physicist with a postgraduate degree in Music Technology. This background connects notational and morphological analysis with computational design; reading older witnesses follows the separate palaeographic assessment protocol. Since the researcher is both a chanter from a specific teaching lineage and editor of the initial lexicon, evaluation independence is secured through three rules: specifications are fixed in numbered versions before each independent assessment; M3a family validation is conducted by other experts, blind to excerpt provenance, with mandatory participation by an annotator outside the editor's lineage; and the researcher's own judgements are recorded exclusively as definition and development material, never as independent confirmation. Sources from the same teaching tradition, such as Karas's tables, are marked as shared sources in the knowledge-role ledger.

5.4 Ethics

Protocols involving human participants (expert annotation, M3c and M3b listening tasks, and new targeted M4 recordings) are submitted to the Research Ethics Committee of the Hellenic Mediterranean University, with the process beginning in WP0. The approved protocol specifies informed consent, pseudonymisation, withdrawal arrangements and data protection procedures. Recordings are used under an existing research-use licence or new explicit consent; the optional neural extension requires separate approval.

6. Timeline, deliverables and publication plan

The 48-month programme separates pilot preparation, finalisation of choices and final evaluation, ensuring that each methodological decision precedes the data used to assess it. Four milestones organise the design. At **month 6**, WP0 is completed and decisions are made on G_{pair}, G_{ATL} and G_{rec0}. At **months 12–18**, historical recovery adequacy is assessed (G_{rec1}), with M_{thesis} identifiability assessed at month 14 (P1). At **month 24**, the validated M3a lexicon is fixed, enabling the M3 stimuli to be finalised. At **month 36**, the main M2 comparison is completed: its models are fixed before delivery of the minimum M4 performance criterion, and calculation of Δ_{hist} is the final step. R4 begins only afterwards. Table 10 and Figure 5 show the same allocation.

The historical and pilot strands have separate time windows and adequacy checks. This allocation protects resources for the main studies, while the status of each quantitative result follows actual data availability.

Table 10. Work packages, months and deliverables

Package	Months	Main work and deliverables
WP0	1–6	Source and provenance registry; first fixed specifications; decisions on G_{pair}, G_{ATL}, G_{rec0} and initial assessment of performance coverage; start of ethics process
WP1	4–12	Common symbolic layer, library with reconstruction checks, first operational EISE pipeline, finalisation of M1 protocol and $g_{\min}$
WP2 (M1)	10–24	$U \times F$ and structural-unit comparisons, selection of $R^{\star}$, formula-level benchmark
WP3 (M3a)	12–24	Independent expert assessment; validated family lexicon
WP4 (A2)	12–36	Assessability of diagnostic relations; 2×2 factorial design
WP4 (M2)	24–36	Matched B0–B1–B2–EISE comparison, supervision curves, Δ_{hist} with uncertainty intervals
WP5 (M3c, M3b)	20–40	Expectation model and stimulus bank (20–28); pilot checks and power simulation (28–30); data collection with the same listeners (30–40)
WP5 (M4)	30–42	Minimum performance criterion by month 36; broader timing and pitch profiles
WP6 (historical route)	1–30	R0 (1–6); R1–R2: digital encoding, candidate classes, synthetic verification (6–20); R3 and ATL transfer test (18–30), subject to adequacy criteria
WP6 (M_{thesis}, P0–P2)	4–22	Relation registry (P0, by 8); identifiability report (P1, by 14); simple models and Δ_{family} (P2, by 22)
WP6 (R4)	36–42	Historical-label accessibility in fixed representations; EISE versus B1 and symbolic input, B2 separately; in the same window, comparison of B1/B2/EISE as M_{thesis} context encoders
WP7 (M5)	36–44	Complete-work maps and selection and realisation profiles for Nikolaos of Smyrna; full M5 subject to adequacy
WP8	42–48	Synthesis, dissertation writing, final documentation of the open repository

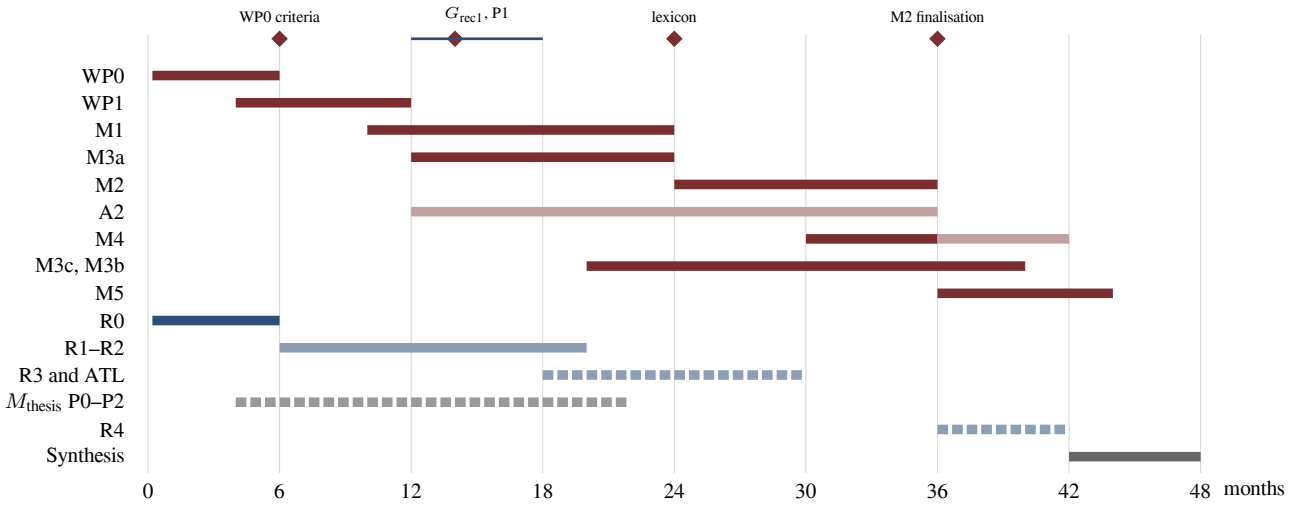


Figure 5. The 48-month sequence. Red lines: Route A; blue: historical route; grey: pilot M_{thesis}. Solid lines denote the core and scheduled deliverables; lighter or dashed lines denote studies with separate adequacy criteria or broader analyses. Diamonds mark milestones; R4 follows M2 finalisation. Detailed allocation appears in Table 10.

Publication plan. Results are organised into distinct outputs, with submission following completion of each study, beginning with those whose value is independent of the confirmatory hypotheses' outcomes. The benchmark, M1 and the relationship between statistical and musicological units target *ISMIR* or the *Transactions of the International Society for Music Information Retrieval* (TISMIR); the pilot M_{thesis} targets *ISMIR* or *Digital Libraries for Musicology* (DLfM). The main M2 study, with supervision curves and the performance criterion, is intended for TISMIR or the *Journal of New Music Research*. ATL, where pairs are sufficient, and ATL-INV, with the aligned corpus of older witnesses and exegeses and the recoverability map, target *ISMIR*, TISMIR or a machine-learning workshop; the historical corpus also targets DLfM. M3c and M3b target a music-cognition journal such as *Music Perception*, while M3a and M5 (formula typology, morphology of exegesis, the Nikolaos of Smyrna study) target international and Greek musicological journals.

7. Risks and predefined alternatives

The main risks concern empirical coverage of historical and performance data and the discriminative reliability of definitions. Each risk has a predefined detection indicator and alternative action (Table 11), so that any design modification is decided before results are known.

Table 11. Risk registry, indicators and predefined responses

Risk	Detection indicator	Predefined response
1. Few verified exegesis pairs	G_{pair} identifies insufficient B2–EISE common support	M2 is restricted to the stricter common support and reported exploratorily; M1, B0–B2 and M3a are completed as planned
2. Insufficient performance coverage	Borderline or inadequate G_{perf} outcome	Analysis at the coarser division; if inadequate, use of the prespecified fallback expert criterion with secondary status; the criterion is fixed before results
3. Low specification reliability	A family fails the two-expert agreement assessment	The family remains a candidate or is withdrawn; disagreement is recorded as a musicological finding; model results cannot generate labels
4. Difficulty recruiting listeners	Sample below that required by power simulation (months 28–30)	Broader recruitment through chant choirs, Byzantine music schools and students; data adequacy for reliable estimation is stated for each hypothesis
5. Access and rights restrictions	WP0 audit of rights and digital-encoding time	Focus on documented available sources; optional optical recognition is disabled if it fails the quality criterion
6. Insufficient ATL pairs	G_{ATL} identifies too few parallel exegeses or families for holding out	A descriptive morphology of transformations is delivered; the transfer test remains inactive
7. Few or conflicting older witnesses	G_{rec0} or G_{rec1} identifies low readability, coverage or witness agreement	R3 delivers a study of available cases, an inventory and documented ambiguities; disagreements are reported as findings
8. Trivial recovery	The sign survives unchanged in the New Method in most passages	Such passages remain as controls; the main set is restricted to distinctions requiring recovery through melodic organisation
9. Low M_{thesis} identifiability	P1 identifies a high proportion of ambiguous cases	The registry and identifiability report are delivered as findings; P2 is estimated only on identifiable cases
10. Mixing synthetic data with evidence	Provenance-field audit before each training run	Synthetic items are excluded from M2 and evaluation sets
11. Low precision of the main comparison	Wide Δ_{hist} uncertainty interval relative to the prespecified meaningful effect	Precision and measurement reliability are reported; the hypothesis remains open within the uncertainty range (§8.5); M1, M3 and M5 are assessed independently

8. Expected outcomes and research contribution

Each contribution introduced in §1.4 is positioned here against its closest precedent, with its evaluation criterion, deliverable and scope of inference (Table 12). Final evaluation distinguishes predictive performance, musicological acceptability, historical documentation and cognitive function.

8.1 C1. Structured formula recognition and benchmark

The closest precedents are local boundary and melodic parallelism models (Cambouropoulos, 2001a, 2006), categorisation of melodic segments through cue abstraction (Cambouropoulos, 2001b), retrieval of known melodic segments (Janssen et al., 2017) and statistical segmentation of liturgical chant (Lanz and Hajič, 2025). C1 connects

them in one task: joint segmentation and family recognition in partially annotated continuous works, with externally defined categories, alternative acceptable boundaries, unknown families and abstention, evaluated through metrics that assess boundaries and labels jointly. The originality lies in the task definition, corpus construction and evaluation design, with a semi-Markov CRF as the estimator. Deliverables comprise the formula-level corpus, including alternative boundaries, hard pairs and the source of every occurrence, together with reference models. The task provides a methodological basis for comparative tests in traditions with documented melodic types.

8.2 C2a. History as a source of supervision

In representation learning, positive pairs usually derive from researcher-selected augmentations or shared category labels (Chen et al., 2020; Khosla et al., 2020), and in music from shared tune families (Karsdorp et al., 2019). The closest precedent, matching the same underlying object for generalisation (Mahajan et al., 2021), obtains this relation from data collection or induces it through the model. M2 derives the relation from independent historical evidence and compares it with shared category membership under matched conditions. Its originality lies in the question and design: supervision provenance becomes an experimental variable, with historically documented pairs and a criterion kept outside training. The question has broader relevance to learning from small but richly documented corpora. A2 separately addresses when diagnostic knowledge helps through its content and when through the structure it adds. Deliverables are Δ_{hist}, supervision curves and diagnostics of the underlying mechanism.

8.3 C2b. Exegesis as a computationally analysable historical channel

Noisy-channel models for back-transliteration, statistical decipherment and reconstruction of language change (Knight and Graehl, 1998; Snyder et al., 2010; Bouchard-Côté et al., 2013), together with computational decipherment of Znamenny notation (Gusev et al., 2020), are the closest methodological precedents. ATL-INV constructs a testable historical channel with three features: reference labels established by blind reading of a specific witness; comparison among candidates with shared metrophony, measuring the additional predictive value of analytical exegesis; and explicit modelling of possible dependence between two exegeses of the same passage. Comparison with known exegetical rules gives the result musicological significance by measuring what the model extracts beyond what theory already encodes. ATL develops melodic-variant matching through fragmentation and consolidation (Mongeau and Sankoff, 1990; Giraud et al., 2014) into a learned probabilistic channel and tests its transfer to new families; R4 measures how much historical information remains accessible in M2 representations.

The pilot M_{thesis} opens a third direction: modelling the relationship between type and realisation. The estimand Δ_{family} measures how strongly family constrains realisation, while typicality is assessed separately from acceptability. Forward prediction of realisation and its inversion towards family complement the historical channel. Extension to neural generation of realisations or joint estimation with historical recovery forms a subsequent research direction, subject to Tier 5 adequacy checks.

The historical strand delivers an inventory and a documented assessment of access, readability and correspondence. Eligible material supports blind readings, alignments and an ambiguity record. When the relevant adequacy checks are passed, calibrated predictions and a quantitative recoverability map are added: which older notational distinctions survive in analytical notation, in which formulae and contexts, and how far recovery exceeds known rules. This map constitutes a quantitative tool for the historical palaeography of chant, within the scope validated by independent readings.

8.4 C3. Musicological and cognitive knowledge

The typology turns historical names and morphological observations into revisable specifications. Classified expert disagreements show where a category is coherent and where it depends on analytical level, modal interpretation or source. The morphology of exegesis records which relations are preserved, where expansions and contractions concentrate, and which changes are associated with exegete, context or notation. With adequate historical evaluation, the recoverability map gives measurable substance to discussions of the great hypostases (Alexandru, 2012) and qualitative distinctions (Karas, 1982), consistently relating findings to actual witness agreement and documented correspondence.

M3c and M3b extend Mavromatis's cognitive question (2005b) and the discussion of categorisation by diagnostic features in relation to overall similarity (Cambouropoulos, 2001b, 2009): they test whether families contribute to melodic expectation and organise relatedness judgements under controlled surface distance, among listeners with different musical experience. In M5, selection, realisation, sequencing and functional role characterise different aspects of Nikolaos of Smyrna's compositions against the declared reference corpus Ω. The distinction between composition through selection and composition through realisation is a specific musicological question, with an inde-

pendent comparison supplied by the published analysis of one of Nikolaos's Communion chants (Ioannidis, 2022). Tradition, written source, contemporary performance and listening are treated as different levels of evidence.

Table 12. Contributions, testable claims, comparators and scope

	Testable claim	Comparator and evidence	Scope of the claim
C1	The explicit formula unit and selected representation support joint segmentation and family recognition in new works, with abstention where evidence is insufficient	Event Transformer, Pitman–Yor, LBDM, parallelism; historical hard cases; held-out works	Value of the formula as a unit in the New Method papadic repertory, within the corpus's scope
C2a	The provenance of positive relations changes the geometry of within-family variation, measurably affecting agreement with performance; diagnostic knowledge helps under specified conditions	B0, B1, B2; perturbed knowledge; independent performance criterion	Effect of the historical provenance of pairs under matched architecture and budget
C2b	Exegesis carries recoverable information about older notation beyond metrophony and rules; transformation transfer to new families and the family's added information for realisation can be measured	Metrophony and context; probabilistic rules; fixed transducer; $q(X \mid C)$; B1 and symbolic input in R4	Written information preserved by exegesis about older notation, and its accessibility in learned representations
C3	Families can be applied by independent experts, organise listening and support distinct characterisation of complete works	Blind expert judgements; listeners with different experience; reference corpus Ω; analysis of Nikolaos's Communion chant (Ioannidis, 2022)	Applicability of the typology, the role of families for participating listeners, and historical characterisation of the works

8.5 Interpreting research outcomes

Interpretation of differences depends on uncertainty, measurement reliability and coverage. Table 13 defines what each outcome supports; an indistinguishable result may leave the question open.

Measurement precision and reliability determine how narrowly the gain under study can be estimated when a difference is indistinguishable. The main deliverables—lexicon, benchmark, representation comparisons, independent judgements and historical inventory—provide the documented basis for all confirmatory analyses.

9. Future transfer to other modal traditions

Transferring the methodology to a second tradition is its next empirical test. This requires locally documented units, independent specifications and historical relations that can be annotated reliably. The shared question would be whether supervision provenance adds information under controlled comparison, and whether analytical notation preserves recoverable traces of concise notation.

A candidate comparative field is Znamenny chant in the Russian Orthodox tradition, also preserved in Old Believer communities around Lake Peipus in Estonia. Its neumatic notation derives from Byzantine notation, its modal organisation is based on the oktoechos, and research on its melodic types (*popevki*) and on analytical elaborations (*razvody*) of melismatic formations written in concise form (*fity*) provides a starting point for an inventory (V. Grigorjeva, 2023; N. P. Parfentiev and N. V. Parfentieva, 2023). A computational precedent also exists in decipherment of older Znamenny notation (V. D. Gusev et al., 2020). Such a test would require expert readers, verification of the link between concise notation and analytical rendering, source access and appropriate collaboration; establishing functional comparability of sources is part of that preparation. The extension belongs to priority tier 5 and lies outside the critical path. Gregorian chant, already studied extensively through computation, remains a methodological reference; possible pretraining on a Cantus corpus would have its own transfer checks.

Selected references

Greek-language sources

Alygizakis, A. E. (1985). *Η οκταηχία στην ελληνική λειτουργική υμνογραφία* [The oktoechos in Greek liturgical hymnography]. Thessaloniki: P. S. Pournaras.

Apostolopoulos, Th. K. (2002). *Ο Απόστολος Κώνστας ο Χίος και η συμβολή του στη θεωρία της μουσικής τέχνης* [Apostolos Konstas of Chios and his contribution to the theory of musical art] (Meletai 4). Athens: Institute of Byzantine Musicology.

Arvanitis, I. (2010). *Ο ρυθμός των εκκλησιαστικών μελών μέσα από την παλαιογραφική έρευνα και την εξήγηση της παλαιάς σημειογραφίας* [Rhythm in ecclesiastical chants through palaeographic research and the exegesis of older notation] (Doctoral dissertation, 2 vols.). Ionian University. https://doi.org/10.12681/eadd/21885

Table 13. Interpretation of possible outcomes under the predefined protocol

Outcome	Scientific significance
EISE outperforms B2 ($\Delta_{\text{hist}} > 0$)	The historical relation yields closer agreement with the independent performance profile over the common eligible support, within the stated uncertainty bounds
Small or indistinguishable Δ_{hist}	Defines the observed gain from historical supervision under the protocol, interpreted with verification checks and material adequacy; information about older notation in the exegesis is tested separately in ATL-INV
B2 outperforms EISE ($\Delta_{\text{hist}} < 0$)	Family supervision agrees more closely with performance variability; separate diagnostics examine whether the finding relates to compression of local differences
The first exegesis improves recovery ($\Delta_{\text{rec}} > 0$)	Analytical notation carries predictively usable information about the witness's notation beyond metrophony and context in the tested works and classes
Additional gain over the comparator ($\Delta_{\text{rec}}^{\text{rule}} > 0$)	The model exploits predictive regularities beyond the preselected strong comparator. Separate comparison with q_{rule} identifies gains over published rules
Small or indistinguishable recovery gain (Δ_{rec} or $\Delta_{\text{rec}}^{\text{rule}} \approx 0$)	The distinction remains unresolved with available data and models, or the added gain over the strongest comparator is small. The result defines recoverability within the examined classes; uncertainty, witness counts, reading agreement and the cause of each unresolved case are reported
The second exegesis adds information ($\Delta_{\text{2ex}} > 0$)	The second exegesis improves prediction on the same paired passages; dependence between exegetes is examined through sensitivity analysis
ATL transformations transfer ($\Delta_{\text{ATL}} > 0$)	Expansion and contraction regularities generalise beyond the estimation families under the declared context
Small or indistinguishable transfer gain	The finding is assessed against coverage, model adequacy and alignment reliability; possible family-specific transformation patterns are examined
Family improves M_{thesis} ($\Delta_{\text{family}} > 0$)	The known family label provides additional predictive value. The role of required constraints and the availability of the label outside the target are reported separately
Small or indistinguishable additional family gain	The measured additional predictive gain from family is small under the particular models and data; identifiability and power checks determine the precision of the inference

Despotis, S. K. (2007). Ερμηνευτικές προσεγγίσεις στο μουσικό υλικό της ελληνικής ψαλτικής τέχνης [Interpretative approaches to the musical material of Greek chant]. *Γρηγόριος ο Παλαμάς*, 818, 417–428.

Ioannidis, Ch. (2022). Η σύνθεση του Νικολάου Πρωτοψάλτου Σμύρνης στο Κοινωνικό «Εἰς πᾶσαν τὴν γῆν» σε ήχο πλ. Δ΄ και ο βαθμός εξάρτησής της από την προγενέστερη παράδοση της εκκλησιαστικής μελοποιίας [Nikolaos Protopsaltes of Smyrna's setting of the Communion chant "Eis pasan ten gen" in plagal fourth mode and its degree of dependence on the earlier tradition of ecclesiastical composition]. *Theologia*, 93(4), 173–193. https://ecclesiagreece.gr/greek/press/theologia/material/2022_4_11_ioannidis.pdf

Karagounis, K. Ch. (2003). *Η Παράδοση και Εξήγηση του Μέλους των Χερουβικών της Βυζαντινής και μετα-Βυζαντινής Μελοποιίας* [The tradition and exegesis of Cherubic Hymn settings in Byzantine and post-Byzantine composition] (Meletai 7). Athens: Institute of Byzantine Musicology.

Karagounis, K. Ch. (2005). *Παραλειπόμενα περὶ τοῦ Χερουβικοῦ Ὕμνου* [Addenda on the Cherubic Hymn]. Volos: Alexandros Xourafas Graphic Arts.

Karas, S. I. (1982). *Μέθοδος της Ελληνικής Μουσικής: Θεωρητικόν* [Method of Greek music: theoretical treatise] (vols. A–B). Athens: Society for the Dissemination of National Music.

Polykarpidis, P. G. (2023). *Υπολογιστική προσέγγιση της Βυζαντινής Μουσικής: Η περίπτωση του Ειρμολογίου του Θεοφάνη Καρύκη* [A computational approach to Byzantine music: the case of Theophanes Karykes's Heirmologion] (Doctoral dissertation). National and Kapodistrian University of Athens.

Spanoudakis, D. P. (2017). *Σύγχρονα μοντέλα ανάλυσης Βυζαντινής Μουσικής: Το φαινόμενο της μελισματικότητας στα κεκραγάρια* [Contemporary models for the analysis of Byzantine music: the phenomenon of melismaticism in the kekragaria] (Doctoral dissertation, 3 vols.). Aristotle University of Thessaloniki. https://doi.org/10.12681/eadd/42193

Stathis, G. Th. (1978). *Η εξήγησις της Παλαιάς Βυζαντινής Σημειογραφίας* [The exegesis of older Byzantine notation] (Meletai 2). Athens: Institute of Byzantine Musicology.

Stathis, G. Th. (1987). Η εξήγηση της ψαλτικής τέχνης [The exegesis of the art of chant]. *Theologia*, 58(2), 337–371.

Toma, I. G. (2018). *Νικόλαος Γεωργίου, πρωτοψάλτης Σμύρνης: Βίος, ψαλτικό έργο, μουσικολογικές προσεγγίσεις* [Nikolaos Georgiou, Protopsaltes of Smyrna: life, chant works and musicological approaches] (Master's thesis). Aristotle University of Thessaloniki.

Chrysanthos of Madytos. (1832). *Θεωρητικὸν Μέγα τῆς Μουσικῆς* [Great theory of music]. Trieste.

Psachos, K. A. (1917/1978). *Η Παρασημαντική της Βυζαντινής Μουσικής* [The notation of Byzantine music] (2nd expanded ed., G. Chatzitheodorou, Ed.). Athens: Dionysos.

Sources in other languages

Alexandru, M. (2012). Some thoughts on the so-called great hypostases of the Middlebyzantine notation. *Musicology Papers*, 27(1), 47–58.

Alexandru, M., & Troelsgård, C. (2020). The "Elements of the Papadikê" and modality features in Byzantine chant. *Series Musicologica Balcanica*, 1(1), 169–204.

Arvanitis, I. (2022). The transcriptions of the Three Teachers: An old or a new sound? *Epistēmēs Metron Logos*, 5, 143–149. https://doi.org/10.12681/eml.28005

Benton, G., Finzi, M., Izmailov, P., & Wilson, A. G. (2020). Learning invariances in neural networks from training data. *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*.

Bouchard-Côté, A., Hall, D., Griffiths, T. L., & Klein, D. (2013). Automated reconstruction of ancient languages using probabilistic models of sound change. *Proceedings of the National Academy of Sciences*, 110(11), 4224–4229. https://doi.org/10.1073/pnas.1204678110

Cambouropoulos, E. (2001a). The Local Boundary Detection Model (LBDM) and its application in the study of expressive timing. *Proceedings of the International Computer Music Conference (ICMC 2001)*. https://hdl.handle.net/2027/spo.bbp2372.2001.021

Cambouropoulos, E. (2001b). Melodic cue abstraction, similarity, and category formation: A formal model. *Music Perception*, 18(3), 347–370. https://doi.org/10.1525/mp.2001.18.3.347

Cambouropoulos, E. (2006). Musical parallelism and melodic segmentation: A computational approach. *Music Perception*, 23(3), 249–268. https://doi.org/10.1525/mp.2006.23.3.249

Cambouropoulos, E. (2009). How similar is similar? *Musicae Scientiae*, 13(1 suppl.), 7–24. https://doi.org/10.1177/10298649090130 0102

Cambouropoulos, E., & Widmer, G. (2000). Automated motivic analysis via melodic clustering. *Journal of New Music Research*, 29(4), 303–317. https://doi.org/10.1080/09298210008565464

Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020). A simple framework for contrastive learning of visual representations. *Proceedings of the 37th International Conference on Machine Learning (ICML)*, PMLR 119, 1597–1607.

Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning phrase representations using RNN encoder–decoder for statistical machine translation. *Proceedings of EMNLP 2014*, 1724–1734. https://doi.org/10.311 5/v1/D14-1179

Chryssochoidis, G., Delviniotis, D., & Kouroupetroglou, G. (2007). A semi-automated tagging methodology for Orthodox ecclesiastic chant acoustic corpora. *Proceedings of the 4th Sound and Music Computing Conference (SMC 2007)*, 126–133.

Conklin, D., & Anagnostopoulou, C. (2006). Segmental pattern discovery in music. *INFORMS Journal on Computing*, 18(3), 285–293.

Conomos, D. E. (Ed.). (1985). *The Treatise of Manuel Chrysaphes, the Lampadarios* (Monumenta Musicae Byzantinae, Corpus Scriptorum de Re Musica 2). Vienna: Verlag der Österreichischen Akademie der Wissenschaften.

Cornelissen, B., Zuidema, W., & Burgoyne, J. A. (2020a). Mode classification and natural units in plainchant. *Proceedings of ISMIR 2020*, 869–875.

Cornelissen, B., Zuidema, W., & Burgoyne, J. A. (2020b). Studying large plainchant corpora using chant21. *Proceedings of DLfM 2020*, 40–44.

De Luca, E., Alexandru, M., & Fujinaga, I. (2023). *Brainstorming on the Encoding of Eastern Neumes: Middle-Byzantine Notation as a Case Study*. TEI Members' Meeting 2023. https://tei-c.org/Vault/MembersMeetings/2023/contributions/128.html

Delviniotis, D., & Kouroupetroglou, G. (2011). DAMASKINOS: The prototype corpus of Greek Orthodox ecclesiastical chant voices. *Proceedings of Crossroads: Greece as an Intercultural Pole of Musical Thought and Creativity*, Thessaloniki.

Di Liberto, G. M., Pelofi, C., Bianco, R., Patel, P., Mehta, A. D., Herrero, J. L., de Cheveigné, A., Shamma, S., & Mesgarani, N. (2020). Cortical encoding of melodic expectations in human temporal cortex. *eLife*, 9, e51784. https://doi.org/10.7554/eLife.51784

Dvořáková, A., Eipert, T., Lacoste, D., & Hajič Jr., J. (2026). Making chant computing easy: CantusCorpus v1.0 and the PyCantus library. *Transactions of the International Society for Music Information Retrieval*, 9(1), 164–178. https://doi.org/10.5334/tismir.321

Gheorghiță, N. (2010). *Byzantine Chant between Constantinople and the Danubian Principalities: Studies in Byzantine Musicology*. Bucharest: Editura Sophia. (Ch. XIV, pp. 225–248: Observations on the technique of transcription (exegesis) into the New Method of the Sunday Koinonikon of the 18th century; also in *Studii şi Cercetări de Istoria Artei*, 4(48), 9–26.)

Giraud, M., Déguernel, K., & Cambouropoulos, E. (2014). Fragmentations with pitch, rhythm and parallelism constraints for variation matching. In *Sound, Music, and Motion: CMMR 2013, Revised Selected Papers* (Lecture Notes in Computer Science 8905, pp. 298–312). Springer. https://doi.org/10.1007/978-3-319-12976-1_19

Gingras, B., Pearce, M. T., Goodchild, M., Dean, R. T., Wiggins, G. A., & McAdams, S. (2016). Linking melodic expectation to expressive performance timing and perceived musical tension. *Journal of Experimental Psychology: Human Perception and Performance*, 42(4), 594–609. https://doi.org/10.1037/xhp0000141

Grigorjeva, V. (2023). Principles of stratification of melodic formulas of Old Russian out-of-ichos chants. *St. Tikhon's University Review. Series V: Christian Art*, 50, 9–23. https://doi.org/10.15382/sturv202350.9-23

Gusev, V. D., Bakhmutova, I. V., Miroshnichenko, L. A., & Titkova, T. N. (2020). Possible approaches to deciphering Russian ancient znamenny chant. *Journal of New Music Research*, 49(3), 298–306. https://doi.org/10.1080/09298215.2020.1762666

Hajič Jr., J., Ballen, G. A., Mühlová, K. H., & Vlhová-Wörner, H. (2023). Towards building a phylogeny of Gregorian chant melodies. *Proceedings of ISMIR 2023*, 571–578.

Hannick, C., & Wolfram, G. (Eds.). (1985). *Gabriel Hieromonachos: Abhandlung über den Kirchengesang* (Monumenta Musicae Byzantinae, Corpus Scriptorum de Re Musica 1). Wien: Verlag der Österreichischen Akademie der Wissenschaften.

Hansen, N. C., Kragness, H. E., Vuust, P., Trainor, L. J., & Pearce, M. T. (2021). Predictive uncertainty underlies auditory boundary perception. *Psychological Science*, 32(9), 1416–1425. https://doi.org/10.1177/0956797621997349

Hariharan, B., & Girshick, R. (2017). Low-shot visual recognition by shrinking and hallucinating features. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*.

Herff, S. A., Harasim, D., Cecchetti, G., Finkensiep, C., & Rohrmeier, M. A. (2021). Hierarchical syntactic structure predicts listeners' sequence completion in music. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 43.

Hewitt, J., & Liang, P. (2019). Designing and interpreting probes with control tasks. *Proceedings of EMNLP-IJCNLP 2019*.

Holzapfel, A. (2010). *Similarity Methods for Computational Ethnomusicology* (Doctoral dissertation). University of Crete. https://doi.or g/10.12681/eadd/30917

Janssen, B., van Kranenburg, P., & Volk, A. (2017). Finding occurrences of melodic segments in folk songs employing symbolic similarity measures. *Journal of New Music Research*, 46(2), 118–134. https://doi.org/10.1080/09298215.2017.1316292

Kaliakatsos-Papakostas, M., Makris, D., Soiledis, K., Tsamis, K.-T., Katsouros, V., & Cambouropoulos, E. (2025). HarmonyTok: Comparing methods for harmony tokenization for machine learning. *Information*, 16(9), 759. https://doi.org/10.3390/info16090759

Kapoor, S., & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9), 100804. https://doi.org/10.1016/j.patter.2023.100804

Karniadakis, G. E., Kevrekidis, I. G., Lu, L., Perdikaris, P., Wang, S., & Yang, L. (2021). Physics-informed machine learning. *Nature Reviews Physics*, 3, 422–440. https://doi.org/10.1038/s42254-021-00314-5

Karsdorp, F., van Kranenburg, P., & Manjavacas, E. (2019). Learning similarity metrics for melody retrieval. *Proceedings of ISMIR 2019*, 478–485.

Karystinaios, E., & Widmer, G. (2024). GraphMuse: A library for symbolic music graph processing. *Proceedings of ISMIR 2024*. https://arxiv.org/abs/2407.12671

Kern, P., Heilbron, M., de Lange, F. P., & Spaak, E. (2022). Cortical activity during naturalistic music listening reflects short-range predictions based on long-term experience. *eLife*, 11, e80935. https://doi.org/10.7554/eLife.80935

Khosla, P., Teterwak, P., Wang, C., Sarna, A., Tian, Y., Isola, P., Maschinot, A., Liu, C., & Krishnan, D. (2020). Supervised contrastive learning. *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 18661–18673.

Knight, K., & Graehl, J. (1998). Machine transliteration. *Computational Linguistics*, 24(4), 599–612.

Koelsch, S., Rohrmeier, M., Torrecuso, R., & Jentschke, S. (2013). Processing of hierarchical syntactic structure in music. *Proceedings of the National Academy of Sciences*, 110(38), 15443–15448. https://doi.org/10.1073/pnas.1300272110

Lanz, V., & Hajič Jr., J. (2025). Gregorian melody, modality, and memory: Segmenting chant with Bayesian nonparametrics. *Proceedings of ISMIR 2025*, 638–646.

Lingas, A. (2003). Performance practice and the politics of transcribing Byzantine chant. *Acta Musicae Byzantinae*, 6, 56–76.

Mahajan, D., Tople, S., & Sharma, A. (2021). Domain generalization using causal matching. *Proceedings of the 38th International Conference on Machine Learning (ICML)*, PMLR 139, 7313–7324.

Mavromatis, P. (2005a). A hidden Markov model of melody production in Greek church chant. *Computing in Musicology*, 14, 93–112.

Mavromatis, P. (2005b). *The Echoi of Modern Greek Church Chant in Written and Oral Transmission: A Computational Model and Its Cognitive Implications* (Doctoral dissertation). Eastman School of Music, University of Rochester.

Mavromatis, P. (2019). Hierarchical structural patterns in improvised music: Implications for cognition [Conference abstract]. *Society for Music Theory, 42nd Annual Meeting, Program and Abstracts*.

Mongeau, M., & Sankoff, D. (1990). Comparison of musical sequences. *Computers and the Humanities*, 24(3), 161–175. https://doi.org/10.1007/BF00117340

Neanes Project. (2026). *Neanes: A Byzantine chant scorewriter* [Open-source software, GPL-3.0]. https://github.com/neanes/neanes

Panteli, M., & Purwins, H. (2013). A quantitative comparison of Chrysanthine theory and performance practice of scale tuning, steps, and prominence of the octoechos in Byzantine chant. *Journal of New Music Research*, 42(3), 205–221.

Papaioannou, C., Benetos, E., & Potamianos, A. (2025). Universal music representations? Evaluating foundation models on world music corpora. *Proceedings of ISMIR 2025*.

Papay, S., Klinger, R., & Padó, S. (2022). Constraining linear-chain CRFs to regular languages. *International Conference on Learning Representations (ICLR)*.

Parfentiev, N. P., & Parfentieva, N. V. (2023). Stroganovs' master of church-singing art Faddey Subotin at the court of Tsar Alexei Mikhailovich. *Journal of Siberian Federal University. Humanities & Social Sciences*, 16(7), 1112–1126.

Pearce, M. T. (2018). Statistical learning and probabilistic prediction in music cognition: Mechanisms of stylistic enculturation. *Annals of the New York Academy of Sciences*, 1423(1), 378–395. https://doi.org/10.1111/nyas.13654

Pearce, M. T., Müllensiefen, D., & Wiggins, G. A. (2010). The role of expectation and probabilistic learning in auditory boundary perception: A model comparison. *Perception*, 39(10), 1365–1389. https://doi.org/10.1068/p6507

Pimentel, T., Valvoda, J., Maudslay, R. H., Zmigrod, R., Williams, A., & Cotterell, R. (2020). Information-theoretic probing for linguistic structure. *Proceedings of ACL 2020*, 4609–4622.

Polykarpidis, P., Kalofonos, D., Balageorgos, D., & Anagnostopoulou, C. (2022). Three related corpora in Middle Byzantine music notation and a preliminary comparative analysis. *Proceedings of ISMIR 2022*, 306–313.

Rabinovitch, G., & Carter-Enyi, A. (2024). Melodic organization and sequential ordering of galant schemata: Implications for eighteenth-century European musicianship. *Music Theory Online*, 30(1). https://doi.org/10.30535/mto.30.1.7

Richard, G., Lostanlen, V., Yang, Y.-H., & Müller, M. (2024). Model-based deep learning for music information research: Leveraging diverse knowledge sources to enhance explainability, controllability, and resource efficiency. *IEEE Signal Processing Magazine*, 41(6), 51–59. https://doi.org/10.1109/MSP.2024.3415569

Ristad, E. S., & Yianilos, P. N. (1998). Learning string-edit distance. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 20(5), 522–532. https://doi.org/10.1109/34.682181

Sarawagi, S., & Cohen, W. W. (2004). Semi-Markov conditional random fields for information extraction. *Advances in Neural Information Processing Systems 17*, 1185–1192.

Savage, P. E., Passmore, S., Chiba, G., Currie, T. E., Suzuki, H., & Atkinson, Q. D. (2022). Sequence alignment of folk song melodies reveals cross-cultural regularities of musical evolution. *Current Biology*, 32(6), 1395–1402.e8. https://doi.org/10.1016/j.cub.2022.01.039

Schindler, A., Herdener, M., & Bartels, A. (2013). Coding of melodic gestalt in human auditory cortex. *Cerebral Cortex*, 23(12), 2987–2993. https://doi.org/10.1093/cercor/bhs289

Snyder, B., Barzilay, R., & Knight, K. (2010). A statistical model for lost language decipherment. *Proceedings of ACL 2010*, 1048–1057.

St. Anthony's Greek Orthodox Monastery. (n.d.). *Papadic Formulae* [Online collection of papadic melodic formulae from 758 compositions]. https://stanthonysmonastery.org/pages/papadic-formulae

Sturm, B. L. (2014). A simple method to determine if a music information retrieval system is a "horse". *IEEE Transactions on Multimedia*, 16(6), 1636–1644. https://doi.org/10.1109/TMM.2014.2330697

Troelsgård, C. (2011). *Byzantine Neumes: A New Introduction to the Middle Byzantine Musical Notation* (Monumenta Musicae Byzantinae, Subsidia 9). Copenhagen: Museum Tusculanum Press.

Tsuboi, Y., Kashima, H., Mori, S., Oda, H., & Matsumoto, Y. (2008). Training conditional random fields using incomplete annotations. *Proceedings of COLING 2008*, 897–904.

Voita, E., & Titov, I. (2020). Information-theoretic probing with minimum description length. *Proceedings of EMNLP 2020*.

Volk, A., & van Kranenburg, P. (2012). Melodic similarity among folk songs: An annotation study on similarity-based categorization in music. *Musicae Scientiae*, 16(3), 317–339. https://doi.org/10.1177/1029864912448329

von Rueden, L., Mayer, S., Beckh, K., Georgiev, B., Giesselbach, S., Heese, R., Kirsch, B., Pfrommer, J., Pick, A., Ramamurthy, R., Walczak, M., Garcke, J., Bauckhage, C., & Schuecker, J. (2023). Informed machine learning: A taxonomy and survey of integrating prior knowledge into learning systems. *IEEE Transactions on Knowledge and Data Engineering*, 35(1), 614–633.

Wu, S., & Cotterell, R. (2019). Exact hard monotonic attention for character-level transduction. *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 1530–1537. https://doi.org/10.18653/v1/P19-1148

Zeng, M., Tan, X., Wang, R., Ju, Z., Qin, T., & Liu, T.-Y. (2021). MusicBERT: Symbolic music understanding with large-scale pre-training. *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, 791–800. https://doi.org/10.18653/v1/2021.findings-acl.70

Candidate's biographical note

Education and professional role. Georgios Spyridakis holds a degree in Physics from Aristotle University of Thessaloniki and a postgraduate degree in Sound and Music Technologies from the Hellenic Mediterranean University. He teaches Physics in public secondary education.

Research experience. His postgraduate research concerned machine-learning classification of musical-instrument vibration patterns from Electronic Speckle Pattern Interferometry (ESPI) images, with training on synthetic data, as well as neural operators for full-field data recovery, an inverse problem. The former experience supports the proposal's computational work; the latter informs the formulation of historical recovery as an inverse problem (§4.8). The extended analysis of the proposal is available in the research repository `github.com/GeorgeSpy/byz-position-phd-study`.

Byzantine music. He has studied Byzantine music since 1998, initially in Thessaloniki. Since 2013 he has studied at the School of Byzantine Music of the Holy Metropolis of Rethymnon and Avlopotamos, from which he received his diploma in Byzantine music, and continues his studies and choir participation under E. Kapaidonakis, in the teaching lineage of Simon Karas and Lykourgos Angelopoulos. His teacher was a founding member of the Greek Byzantine Choir. His extensive chanting practice gives him direct familiarity with the notation, modal system and formulaic phraseology. For this reason, his own judgements are used solely as definition and development material, while other experts undertake independent validation (§5.3).